



GE Medical Systems

Technical Publications

Direction 2289435

Revision 3

ACGD/PDU Upgrade Installation from SGD & RF/PDU

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

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Rev. 11/15/2000



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
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- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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注意：

- 本维修手册仅存有英文本。
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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Rev A . . .	Dec 5, 2000 . . .	Preliminary Release
Rev 0 . .	Jan 12, 2001 . . .	Initial Release
Rev 1 . .	Feb 23, 2001 . . .	Added instructions for installing new Dual Temperature Sensor Cables (Run 749) and SRI board.
Rev 2 . . .	Apr 3, 2001 . . .	Changed adhesive from RTV to epoxy with applicator. Updated input voltage DIP switch setting information.
Rev 3 . . .	Sep 6, 2001 . . .	Added M3090KC and respective instructions for upgrading a CRM. Added instruction for checking fan rotation on ACGD/PDU cabinet (CQA MRIge66116).

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SECTION 1 – OVERVIEW



PRIOR TO UPGRADE!

NEED TO DETERMINE THAT FACILITY INPUT POWER MEETS REQUIREMENTS FOR ACGD CONFIGURED SYSTEM. REFER TO 2223170, SECTION 5, POWER REQUIREMENTS.

1-1 SIGNA HORIZON SYSTEM CONFIGURATION

This manual provides instructions to upgrade a Signa WideOpen system with SGD and RF/PDU cabinets to ACGD/PDU and SRF cabinets. This installation manual is applicable to Fixed-Sites at magnet field strengths of 1.0T or 1.5T.

This Direction covers the following catalogs:

M3090BB – ACGD Upgrade for Systems with SGD and RF/PDU

M3090KC – ACGD Upgrade for CV/i and MR/i SGD Systems

- This upgrade replaces:
 - RF/PDU and SGD gradient cabinets with ACGD/PDU and SRF cabinets.
 - **(For BRM sites only)** Existing Temperature Sensor cable and SRI circuit board with new Dual Temperature Sensor cable and SRI circuit board (P/N 2287391).

The new cabinets include the gradient driver and PDU combined in the ACGD/PDU cabinet and an empty SRF cabinet that will be populated with the existing RF components. The instructions are written assuming that the new ACGD/PDU cabinet will be located close to where the existing RF/PDU cabinet. Thus the ACGD/PDU cabinet will be in close proximity to the incoming power feeds and should minimize power cabling impacts. Before beginning the upgrade, determine if the gradient output cables (Runs 762, 763, and 764) will reach the planned location for the ACGD/PDU cabinet. If not, you will have to procure new set of gradient output cables (2120963–5).

- Connection of the incoming power feeds to the new ACGD/PDU may require an Electrical Contractor to make the conduit connections. Please plan accordingly to have the contractor available if needed.
- Typical installation and completion of this upgrade requires 2 people for 18 hours (BRM) or 14 hours (CRM) each.
- The heaviest of the RF components (RF amps) that are transferred to the new SRF cabinet weigh 34 kg. (75 lbs.). If you desire, plan to have the Universal Hoist Kit (46–317266G5) on hand for these components.
- Eddy Current Compensation calibration must be done as part of this upgrade. Ensure you have an Grafidy or ECMT Kit on hand.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2223751, Signa MR/i & CV/i Upgrade Pre-installation*, Pre-installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon system are listed in *Direction 2223751, Signa MR/i & CV/i Upgrade Pre-installation*, Section 10, TOOLS AND TEST EQUIPMENT.

1-3 PRODUCT LOCATOR

PLC (Product Locator System) tracks creation, shipment, trans-shipment, and installation location of serialized models.

The installation card is not a “warranty card”, therefore it is to be processed as soon as the installation starts.

The PLC system also includes Remanufacture, Scrap, and Transfer cards. They should be obtained and completed for all serialized cabinets removed from the system according to Policy and Procedure.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. Process just the installation card and discard any extra shipment cards and labels.

System upgrade serial-number rating plates are furnished. They provide the means to track systems with various hardware versions of the System Upgrade. Kit added rating plates are furnished with various upgrade kits. They signify that model numbers on the major component have changed function and compatibility and have become a “look-alike” for current production in performance and compatibility. These rating plates do not have shipment cards since serial numbers are not assigned.

1-4 DISCARDED MATERIAL RETURN POLICY

The “discard” or “return pile” consists of items removed by the Signa ACGD Upgrade or traded in for new options and have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the return pile is per GE Medical Systems Policy and Procedures.

1. For coordination of return pile, contact GEMS Recycling Center: Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414–747–6855); WizMail “RECYCLING”; Field ASPEN 1–800–525–1516, Box #26041.

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
 - Site name, address, and phone number
 - Date of completed ACGD upgrade.
 - Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.
2. Prepare the return pile for shipment as follows:
 3. Process product locator cards for all serialized components removed from the system according to policy.
 4. Regulations governing the disposal of this material in the recycle pile vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the “recycle pile” per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-5 UPGRADE SEQUENCE

Follow the system Upgrade Flowchart in SECTION 2 for best installation efficiency.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify facility input power to meet ACGD requirements will be available at time of upgrade installation.
- Verify that the upgrade has arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete cabinet(s) and hardware from equipment room.
- Modify existing cables
- Bring in the new cabinets
- Transfer modules from RF/PDU to SRF cabinet
- **(BRM only, not for CRM)** Remove enclosure covers, bridge, and front end bell for installation of new temperature sensor cable
- **(BRM only, not for CRM)** Remove existing temperature sensor cable and install new cable
- **(BRM only, not for CRM)** Replace existing SRI circuit board with new board
- **(BRM only, not for CRM)** Install front end bell, bridge, and enclosure covers
- Route and connect new cables in equipment room
- Connect all other cables
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1-6 SYSTEM POWER SHUTDOWN

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance HighSpeed line measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.



ELECTRIC SHOCK HAZARD!! HIGH VOLTAGES ARE PRESENT ON INTERCONNECTING CABLES USED WITH SPECIAL FIELD INSTALLED ION-PUMPS USED WITH OXFORD MAGNETS. THE ION PUMP POWER SUPPLY IS USUALLY CONNECTED TO A POWER SOURCE THAT IS NOT CONTROLLED BY THE PDU AND HIGH VOLTAGE WILL BE PRESENT EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER TO THE ION PUMP POWER SUPPLY, LOCK OUT, AND TAG BEFORE PERFORMING PROCEDURES IN THE FOLLOWING SECTIONS. ALSO CHECK FOR OTHER INTERCONNECTING EQUIPMENT WHICH WAS FIELD INSTALLED AND DISCONNECT POWER, LOCK OUT, AND TAG.

4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring.

1-7 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Press the FULL ON button on the PDU front control panel.
4. Except for the ACGD/PDU Cabinet gradient power modules, switch all cabinet circuit breakers to ON. The ACGD power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.

1-8 SOFTWARE LOAD FOR ACGD UPGRADE

Perform the applicable Load from Cold procedure for the software shipped with the upgrade. The procedures are located on the Service Methods CD at:

- **BRM Systems** – Operator Workspace: Setup & Calibration, Signa 9.0 and ASP2 Software Load (LFC).
- **CRM Systems** – Operator Workspace: Setup & Calibration, Host Software Install (Release CNV3 & CNV4)

1-9 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locator installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

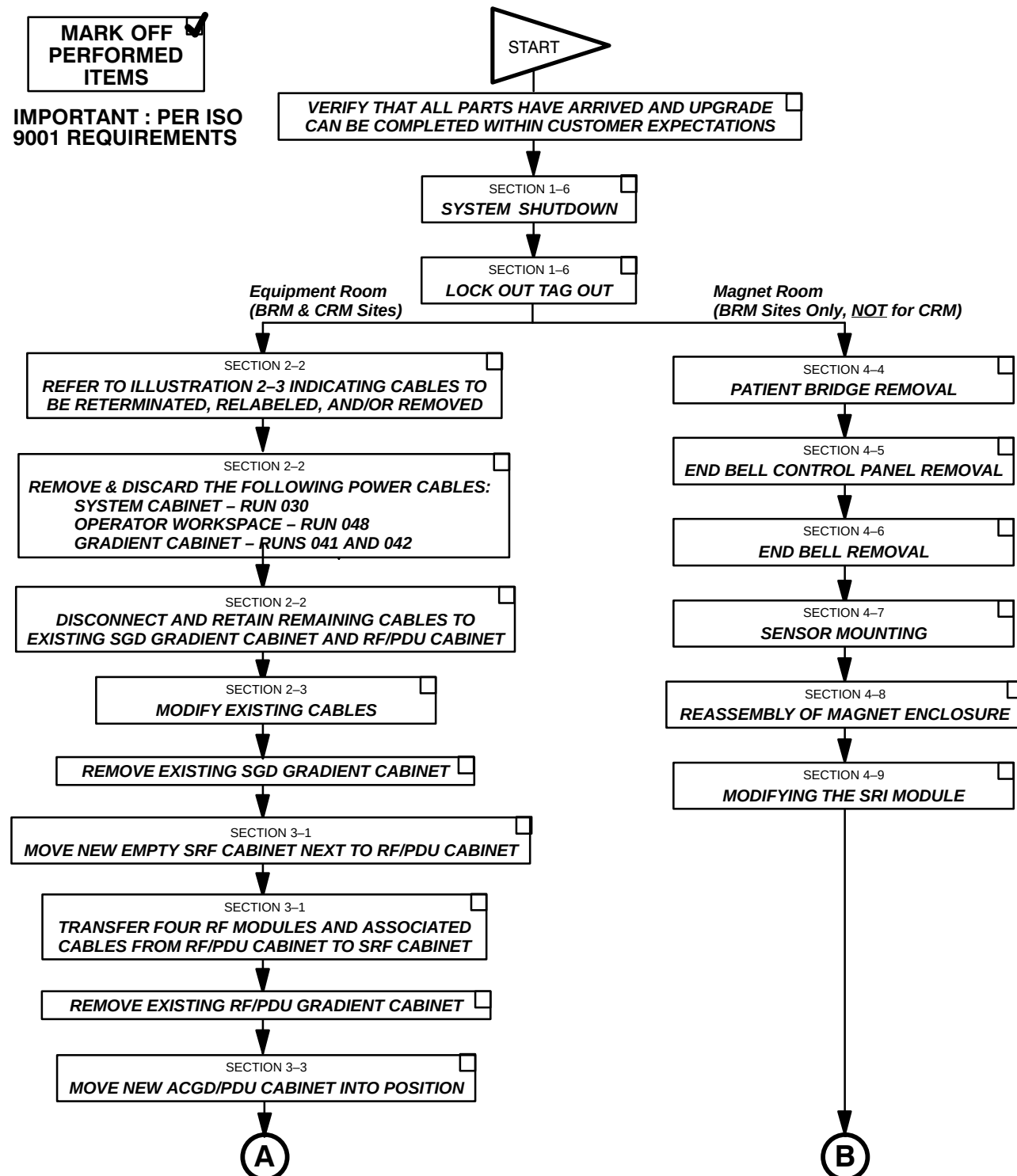
**Product Locator File
P.O. Box 414, W-523
Milwaukee, WI 53201-0414**

Note

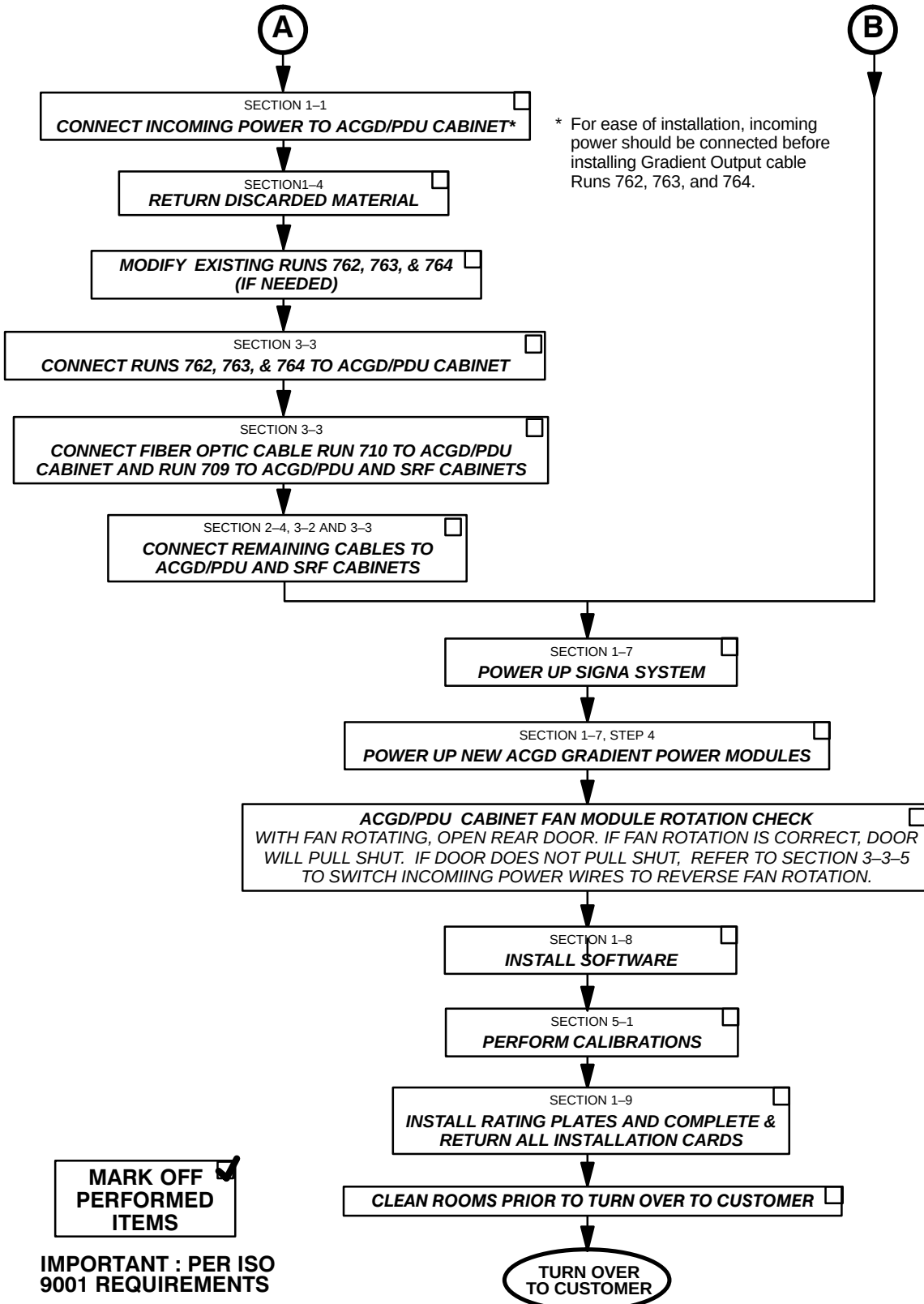
Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

SECTION 2 – FLOWCHART AND CABLE MAPS

2-1 INSTALLATION FLOWCHART



2-1 INSTALLATION FLOWCHART (cont'd)

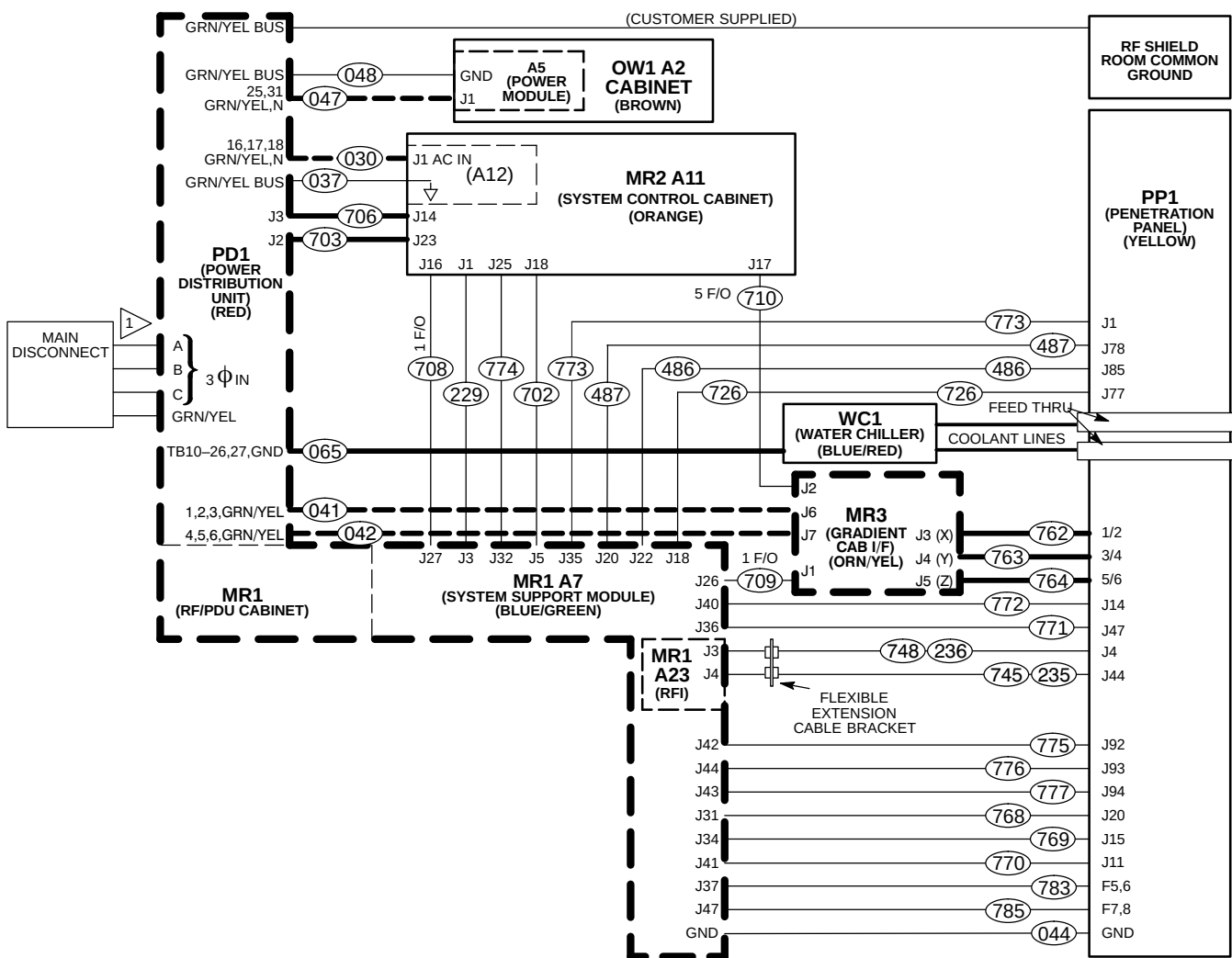
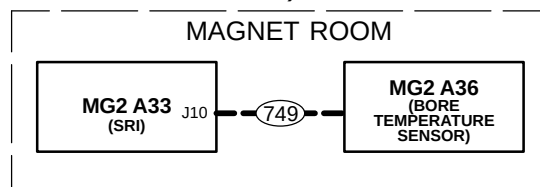


2-2 CABLE MAP FOR CABLES TO BE REMOVED, RETERMINATED, AND/OR RELABELED

LEGEND

- REMOVED CABLES
- ===== RELABELED CABLES
- ===== CABLES TO BE DISCONNECTED FROM CABINETS THAT ARE BEING REMOVED AND TO BE RECONNECTED TO NEW CABINETS.
- [] CABINETS TO BE REMOVED FOR UPGRADE
- 1 CUSTOMER SUPPLIED. POWER FEEDERS MAY NEED TO BE REPLACED TO MEET ACGD POWER REQUIREMENTS.

(Run 749 Cable Installed for BRM Sites Only,
NOT APPLICABLE for CRM)



AFFECTED CABLES WHEN UPGRADING FROM RF/PDU & SGD TO SRF & ACGD/PDU CABINETS

ILLUSTRATION 2-3

2-3 NEW AND RELABELED CABLES

Several existing SGD 8.x cables need to be re-labeled and then re-connected per the new labels. Refer to Table 2-1.

TABLE 2-1
SGD 8.x CABLES RE-LABELED AND RE-CONNECTED FOR UPGRADE

OLD RUN	ONE END		CONNECTS TO	OTHER END	
	OLD LABEL	NEW LABEL		OLD LABEL	NEW LABEL
065	RUN #065 PD1 2223355	RUN #970 MR3-PD1 2280289	16, 17, GND	RUN #065 WC1 2223355	
703	RUN #703 PD1-A1-A2-J2 PD1-A14-A1-J2 46-258431G809	RUN #703/971 PD1-A1-A2-J2 PD1-A14-A1-J2 MR3-PD1-J2	MR3-PD1-J2	RUN #703 MR2-A11-J23 46-228431G809	RUN #703/971 MR2-A11-J23 46-258431G941
706	RUN #706 MR1-PD1-J3 PD1-A14-A1-J3 46-258431G808	RUN #706/972 PD1-A1-A3-J2 PD1-A14-A1-J3 MR3-PD1-J3	MR3-PD1-J3	RUN #706 MR2-A11-J14 46-258431G808	RUN #706/972 MR2-A11-J14 46-258431G940

New cables to be installed for this upgrade are indicated in Table 2-2.

TABLE 2-2
NEW CABLES TO BE INSTALLED

RUN	PART NUMBER	FROM	TO
038	46-255947G608	MR3-PD1	MR1-A7-GND
749*	2285729	MG2 A33 J10	MG2 A36
966	2280286	MR3-PD1-7,8,9,GND	MR1-A7-J1
967	2280287	MR3-PD1-10,11,12,GND	MR1-CB1
968	2280286-2	MR3-PD1-4,5,6,GND	MR2-A12-J1
969	2280288	MR3-PD1-18,19,GND,N	OW1-A2-A5-J1

* Run 749 Cable Installed for BRM Sites Only, **NOT APPLICABLE** for CRM.

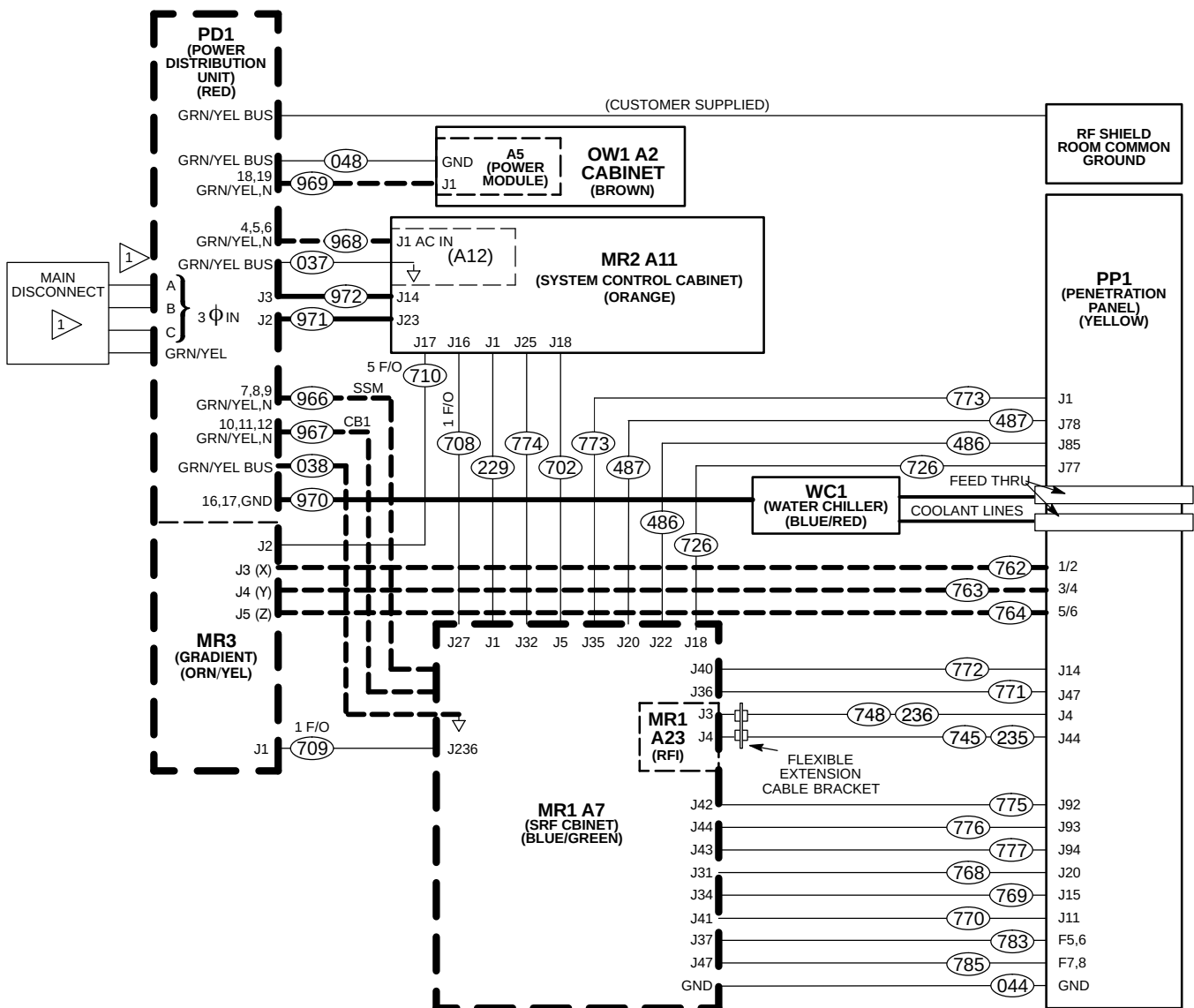
2-4 INTERCONNECT MAP AFTER UPGRADE

Verify that Runs 762, 763, and 764 have sufficient length to route from ACGD/PDU Cabinet to Penetration Panel.

LEGEND

- NEW CABLES
- RELABELED AND RECONNECTED CABLES
- RECONNECTED CABLES TO NEW CABINETS.
- []** NEW CABINET INSTALLED FOR UPGRADE
- 1** CUSTOMER SUPPLIED. POWER FEEDERS MAY NEED TO BE REPLACED TO MEET ACGD POWER REQUIREMENTS.

(Run 749 Cable Installed for BRM Sites Only,
NOT APPLICABLE for CRM)



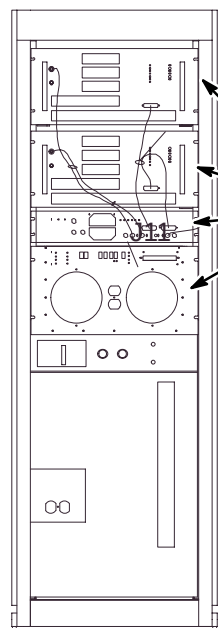
CABLES AFTER UPGRADING TO ACGD/PDU & SRF

ILLUSTRATION 2-4

SECTION 3 – UPGRADING CABINETS

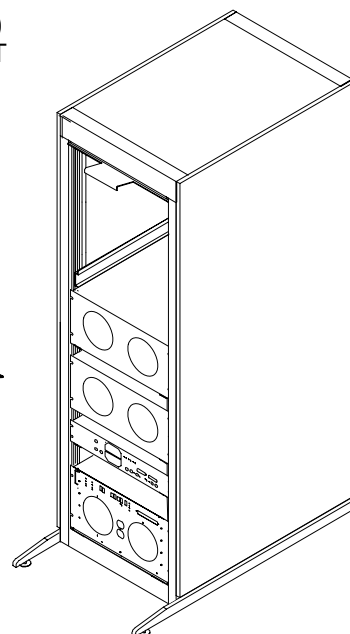
3-1 TRANSFER MODULES FROM RF/PDU CABINET TO SRF CABINET

3-1-1 Transfer of Modules

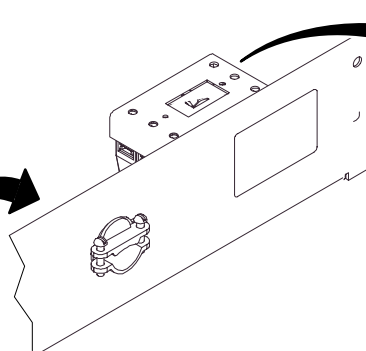
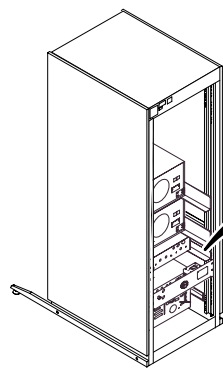


Note: The 1.5T RF/PDU and SRF (Scaleable RF) cabinets are shown in these procedures. The 1.0T uses the same location for these modules

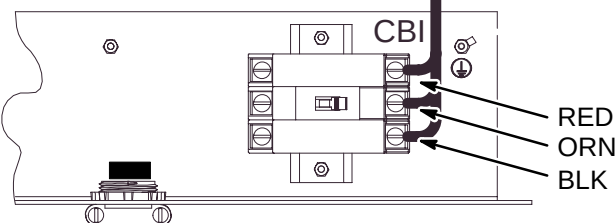
- ① Using at least two people or a hoist device attached to the top of the cabinet, remove the top four modules and associated cables for transfer to empty SRF Cabinet.
- ② Install modules in new SRF Cabinet on preinstalled slides and brackets.
- ③ Reconnect cables.



3-1-2 Connect Power Cord to CB1



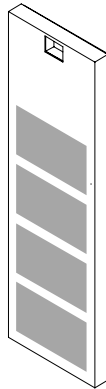
- ① Route and connect power cord from J21 on SRI Module to CB1 as marked.



3-2 INSTALL SRF CABINET

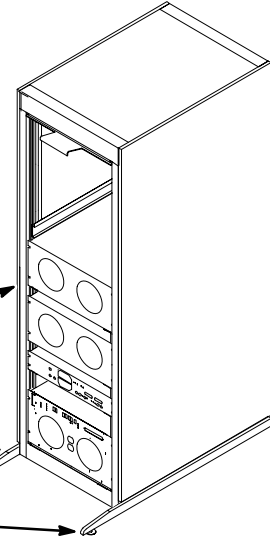
3-2-1 Position SRF Cabinet

Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.

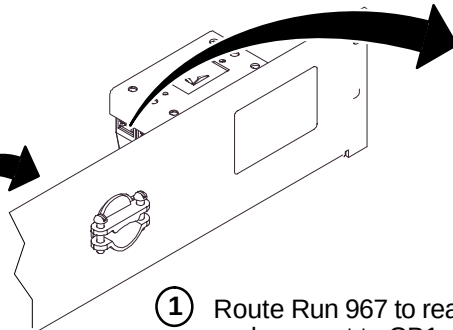
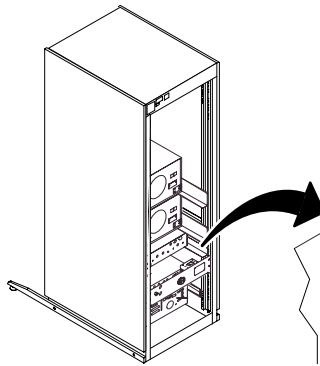


Note: The 1.5T SRF (Scaleable RF) cabinet is shown in these procedures. The 1.0T is the same except for the Heliax cables installed in Procedure C below.

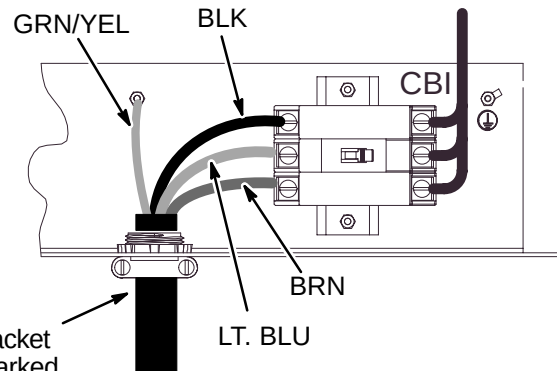
- ① Move new SRF Cabinet to final position.
- ② Remove anti-tip bars from inside of cabinet and attach to each side.
- ③ Adjust leveling pads to level cabinet.



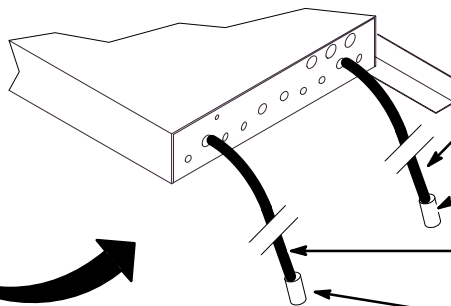
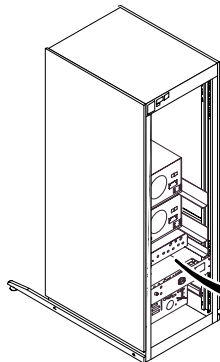
3-2-2 Route/Connect Run 967



- ① Route Run 967 to rear bracket and connect to CB1 as marked.



3-2-3 Connect Runs 235/745 and 236/748 to RF Interface (RFI) Module

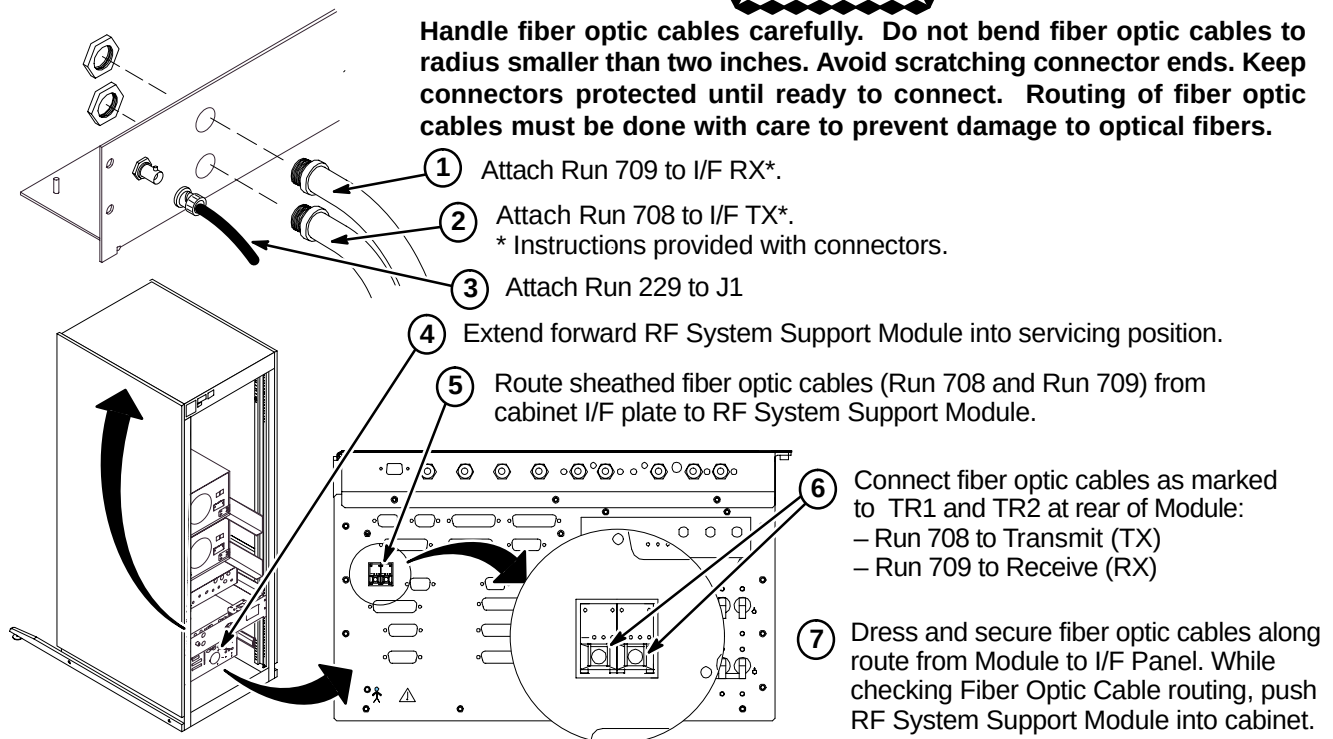


- ① Connect Extension Cable with HN plugs (2105362) to J3 Head Output on RFI Module.
- ② Connect Run 236 (1.0T) or Run 748 (1.5T) to above Extension cable.
- ③ Connect Extension Cable with N plugs (2105358) to J4 Body Output on RFI Module.
- ④ Connect Run 235 (1.0T) or Run 745 (1.5T) to above Extension cable.

3-2-4 Route/Connect Fiber Optics and Run 229

CAUTION

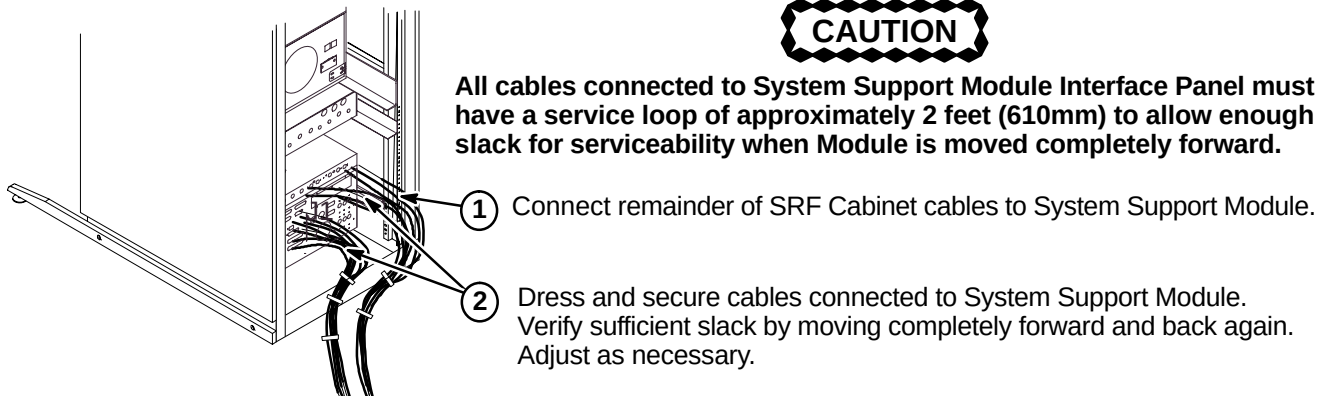
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



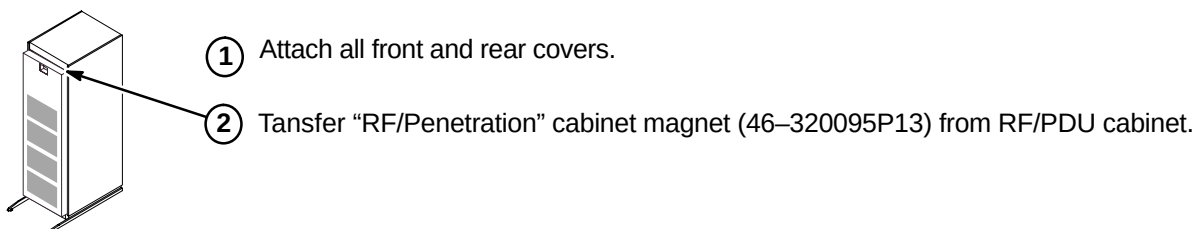
3-2-5 System Support Module Connections

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



3-2-6 Attach InSite Cabinet Magnet



3-3 INSTALL ACGD/PDU CABINET

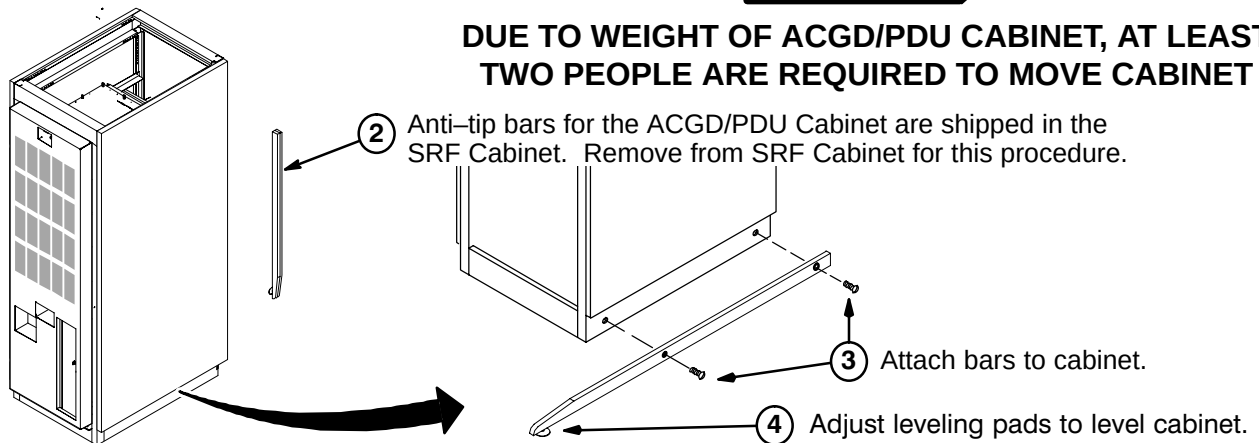
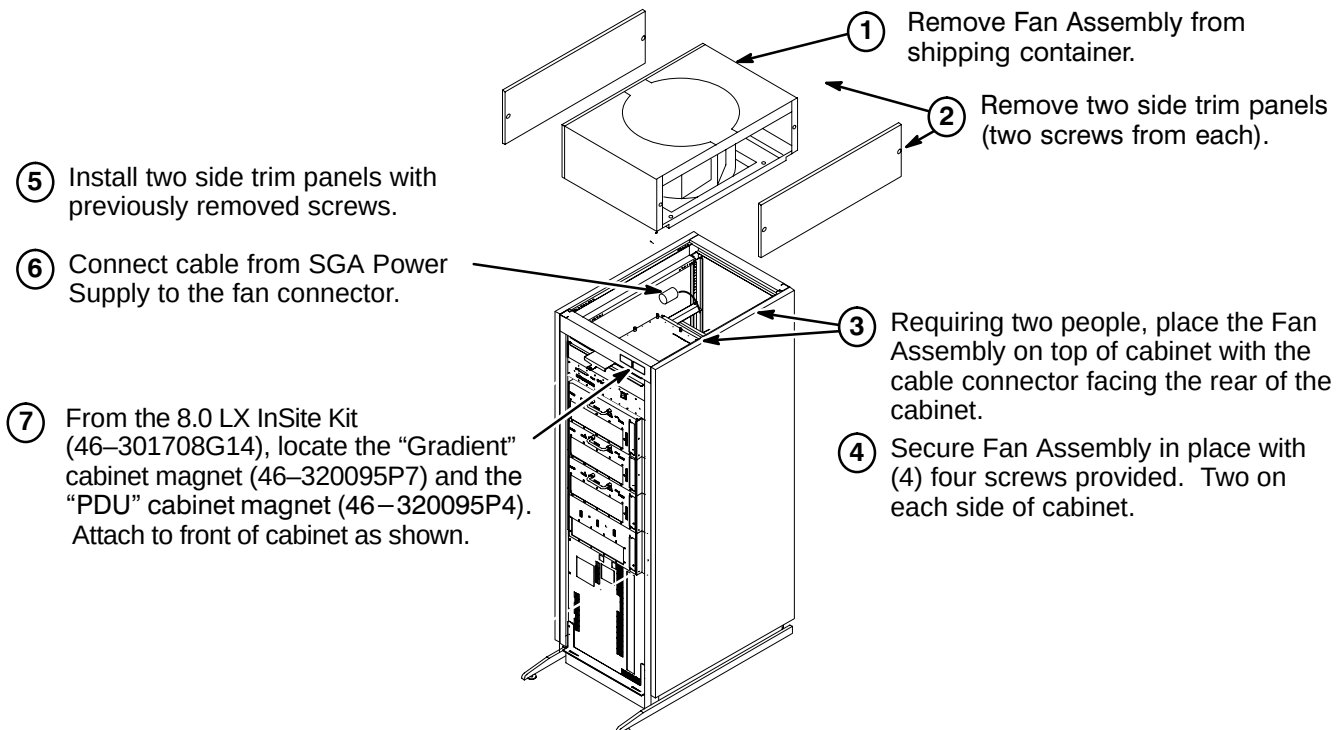
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

3-3-1 Position ACGD/PDU Cabinet

- ① Move cabinet to final position.

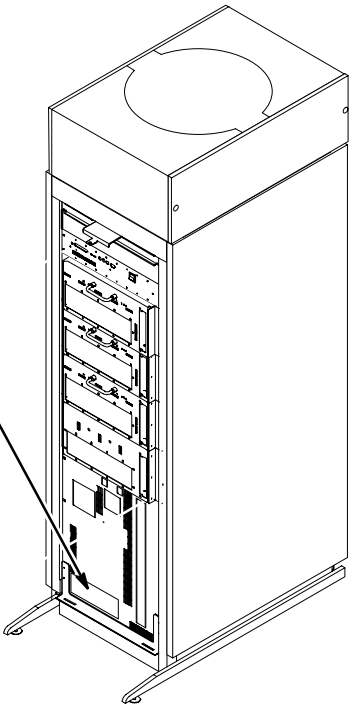
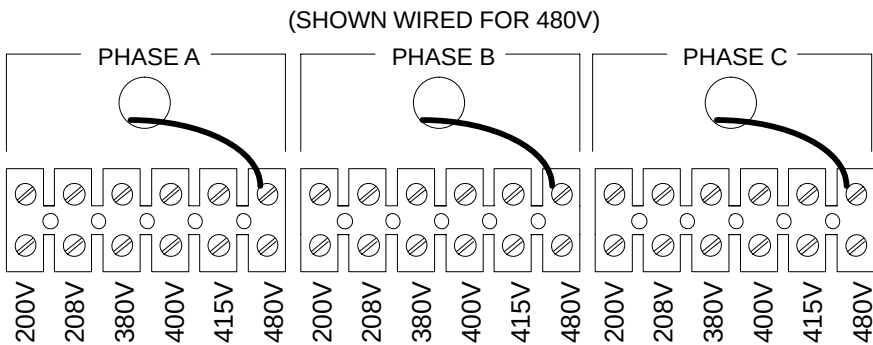
WARNING!

DUE TO WEIGHT OF ACGD/PDU CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

**3-3-2 Install Fan Assembly and Attach InSite Identification Magnet**

3-3-3 Input Voltage Selection

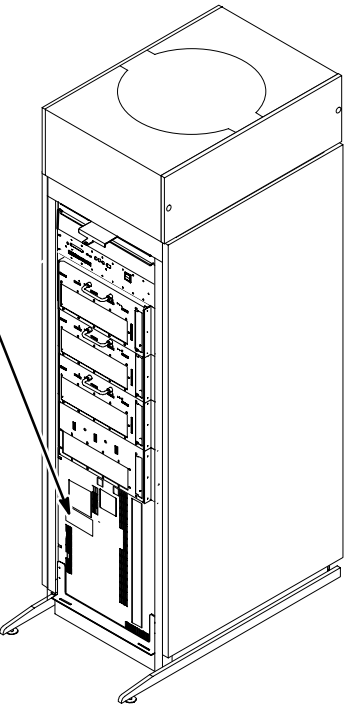
- ① Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- ② Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
- ③ For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



3-3-4 Circuit Breaker Dip Switch Settings

- ① Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
- ② Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter “L” need to be set according to the Input Voltage indicated at the left.

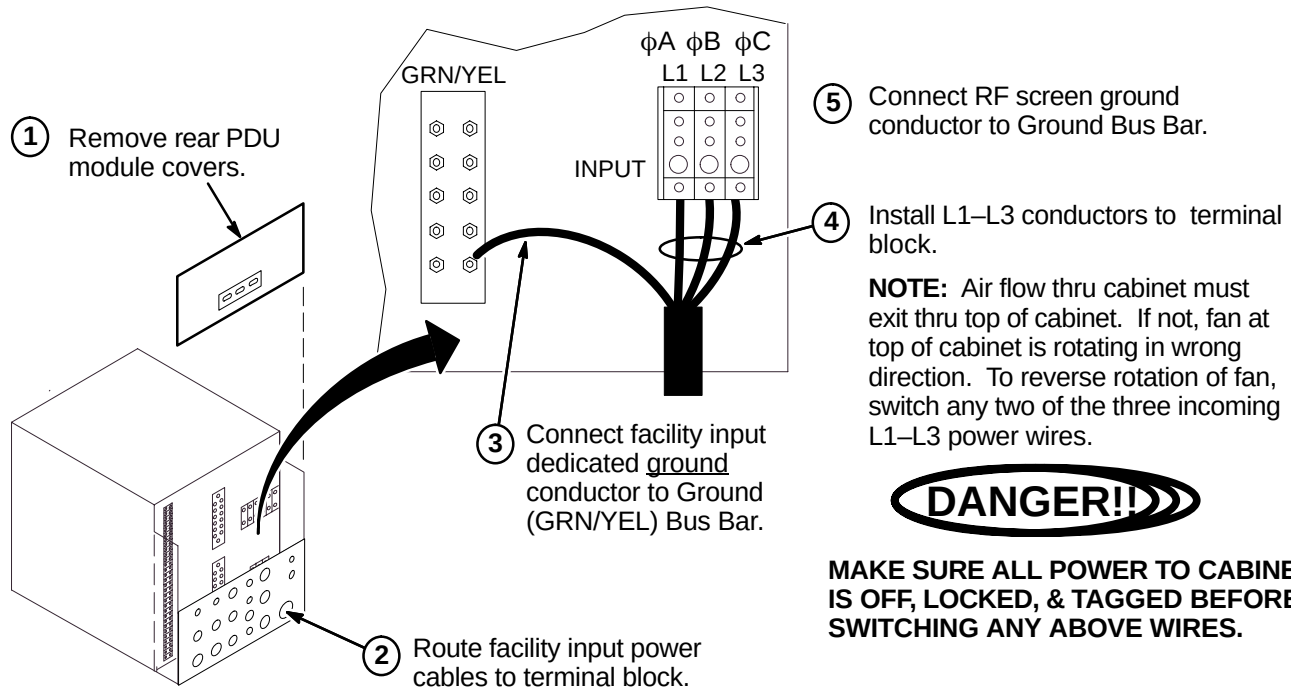
INPUT VOLTAGE	DIP SWITCH “L” SETTING	L	I
200V			
208V		Set these switches according to the input voltage shown at the left.	Set these switches in the UP position as shown.
380V			
400V			
415V			
480V		If these switches are present, leave both of them in the UP position as shown.	



3-3-5 Facility Power and Ground Connection

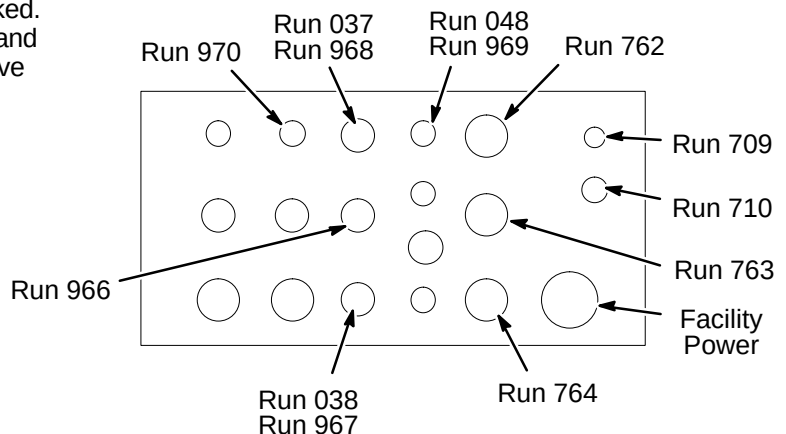
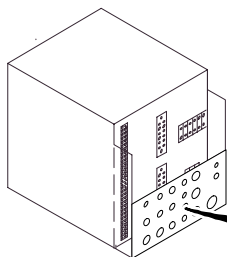
WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.



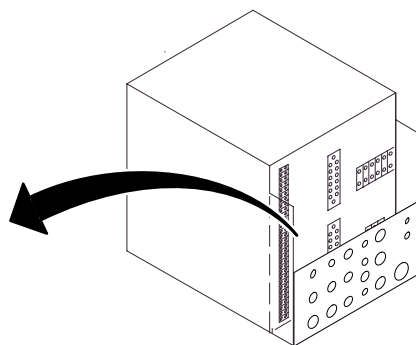
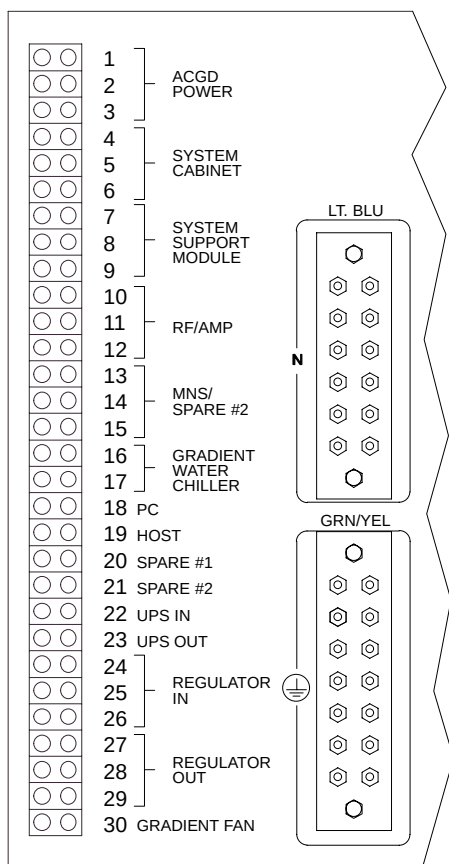
3-3-6 Route Power Cables Thru Interface Panel

- 1 Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 048) **MUST** be routed with their respective power cable Runs thru the same port.
- 2 Route any option Runs as marked.



3-3-7 Run 967 Power Cable Alteration

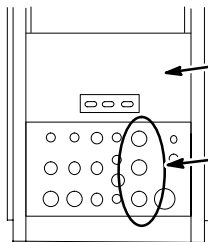
- ① Cut off ring terminal from ORN, RED, and BLK wires.
 - ② Strip 0.75 IN (19mm) of insulation from end of each wire.
 - ③ Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.
-

3-3-8 Connect Power Cables to PDU Module

- ① Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
- ② Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
- ③ Tighten clamps of interface panel to secure cables in place.

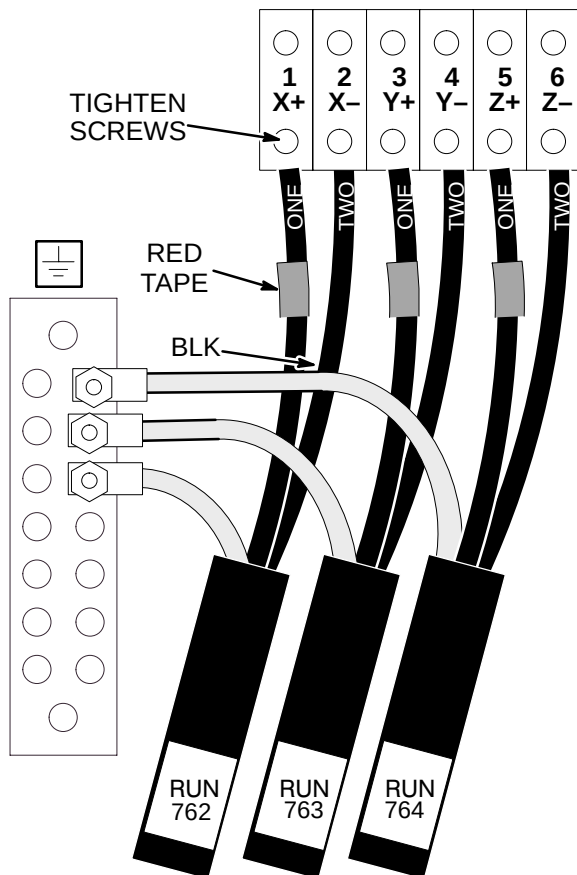
CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	System Cabinet 3 ϕ , Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3 ϕ , Ground
10,11,12	967	ϕ A, B, C	RFI/AMP 3 ϕ , Ground
16,17	970	ϕ A,B	Water Chiller Single ϕ , Ground
18,19	969	ϕ C,A	Operator Workspace Single ϕ , Neutral, GND
20	973	ϕ A	Heat Exchanger Single ϕ , Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire

COLOR CODE: for remainder of cables:
 3 ϕ = Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
 Single ϕ = Brown, Black

3-3-9 Gradient Output Cable Interface PaneACGD CABINET
(REAR VIEW)

NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

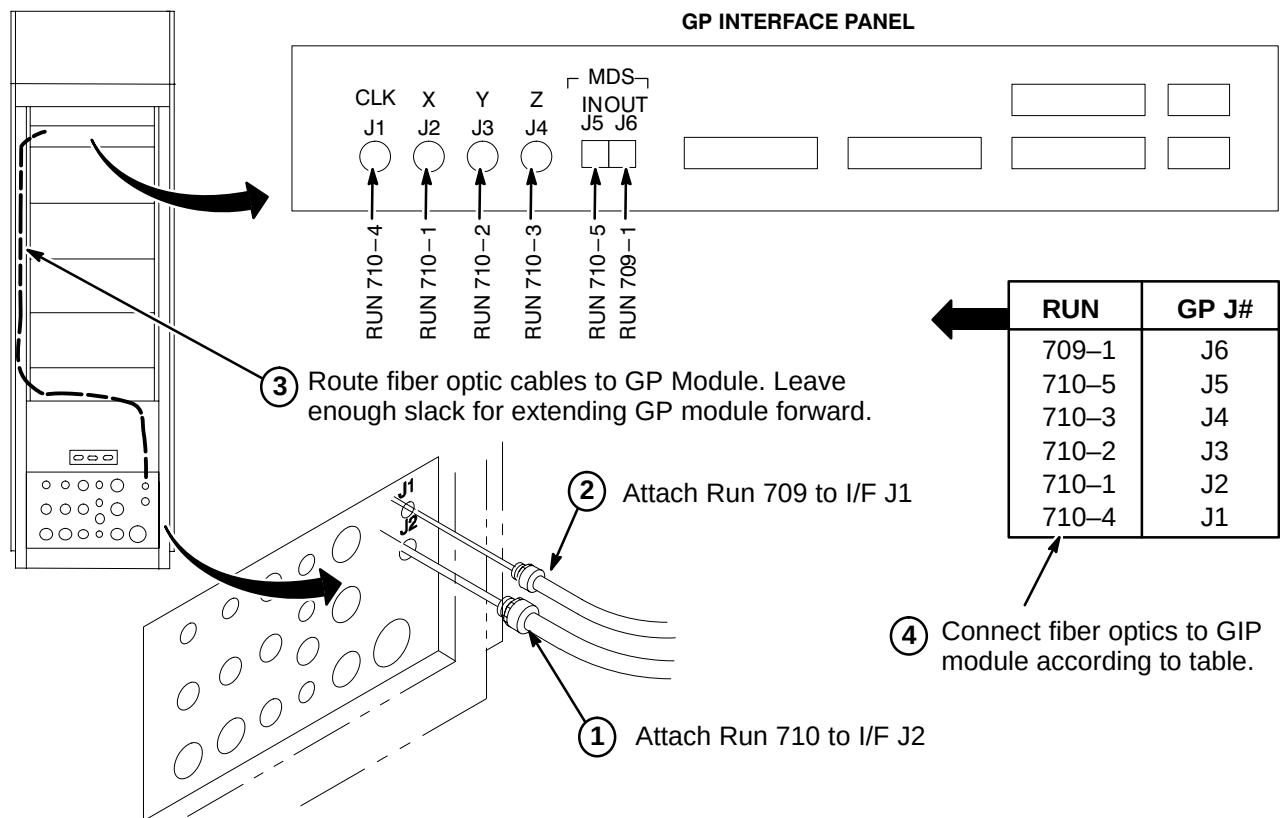
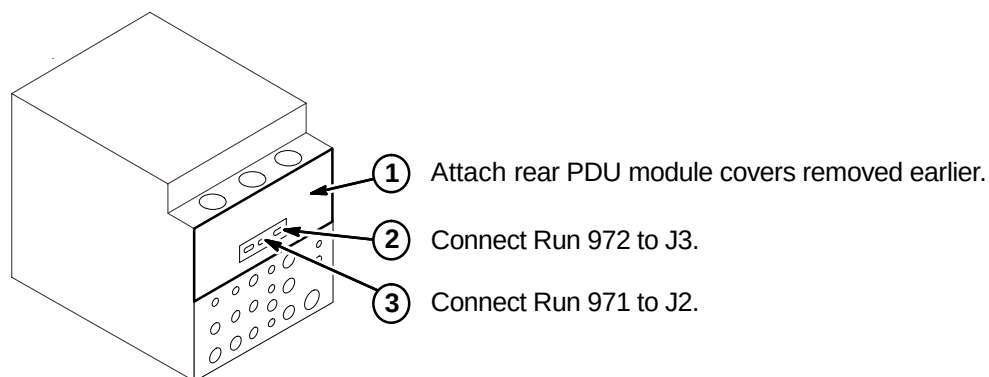
- ① Remove Rear Cover Plate by removing seven screws that hold it in place.
- ② Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

3-3-10 Gradient Output Cable Routing**GRADIENT CABLE CONNECTIONS**

- ① Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.

NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

- ② Tighten screws to 52-65 inch-lbs (5.88-7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
- ③ Connect the GND leads of each run to the Ground Bus Bar studs.
- ④ Tighten clamps on Interface Panel to secure gradient output cables.

3-3-11 Route/Connect Fiber Optics to GP Module**3-3-12 Connect Runs 971 and 972 Cables to PDU Module**

SECTION 4 – THERMAL SENSOR UPGRADE (BRM Systems Only)

Note

This section does not apply to systems with CRM Body Coil.

4-1 INTRODUCTION

The Thermal Sensor Upgrade, #2287391, adds a redundant bore temperature sensor required for safety. This update applies to the Signa WideOpen (LCC) magnet enclosure only.

4-2 TOOLS AND EQUIPMENT

The following tools and equipment are required to perform this procedure:

- Drill bit, 7 mm or 5/16 inch
- Drill
- Dykes
- Digital Multimeter
- Lockout/Tagout equipment
- Chisel, ½ inch (suggested)
- Tape, electrical or duct
- Socket-head capscrew, 6 mm
- Hammer
- Permanent marker, fine-tip
- Slot-head screwdriver, 5/16" blade
- Nut driver, 5/16"
- Static-grounding wrist strap
- Copper tape (for replacement only)

Manpower Requirements: Two or three people are required to place board under the bridge and remove end bell.

The following items are provided in the Thermal Sensor Upgrade Kit, P/N 2287391

- | | | |
|----|-------------|---|
| 1. | 2285729 | Run 749, Dual-head sensor cable |
| 2. | 46-288016G9 | SRI board with dual-sensor daughter board |
| 3. | 46-208747P1 | Cable ties |
| | 46-208747P2 | |
| 4. | 46-282264P1 | Epoxy adhesive |
| 5. | 2298033 | Applicator |
| 6. | 46-320187P1 | Nozzle |

4-3 POWER DOWN PROCEDURE

If complete system is required to be powered down, refer to Section 1-6, SYSTEM POWER SHUTDOWN.

1. Press the **[In Fast]** button on the magnet enclosure to move the carriage all the way to the rear of the magnet.



2. Click the (Service Tools) icon. From the Service Desktop Manager, click  , then  . The system will begin the shutdown process. After about 30 seconds, the CPU will shut down.



3. On the system PC, click  , **[Shut Down...]**, **[Yes]**. The PC will shut down.

Lock out and tag out RF/PDU Cabinet circuit breaker MR1 A20 CB4 at rear of cabinet on System Support Module (SSM).

- Verify that voltage between MR1 A20 TP4 (+) (SRI+24V) and TP13 (-) (SRI RTN) is 0 Vdc.
- Verify voltage between MR1 A20 TP1 (+) (Dock) and TP14 (-) (Dock RTN) is 0 Vdc.

4-4 PATIENT BRIDGE REMOVAL

1. Undock patient transport. Move patient transport out of the way to allow enough room for removal of the magnet end bell.
2. Remove rear pedestal covers.
3. Lift the tension arm to release tension from carriage drive belt (see Illustration 4-1).

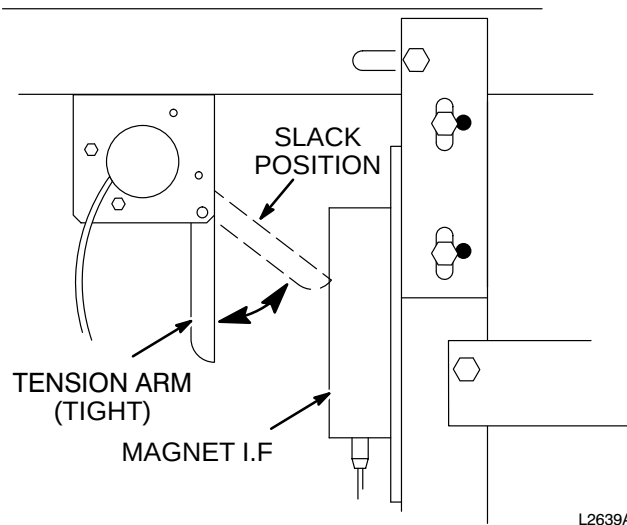
**RIGHT SIDE VIEW OF REAR PEDESTAL (As Viewed From Rear)**

ILLUSTRATION 4-1



Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.

4-4 PATIENT BRIDGE REMOVAL (Cont'd)

4. Use an allen wrench to remove four M6 bolts holding down bridge. Two of the bolts are in the front of the bore and two bolts are in the back of the bore. Lift front of bridge off front bridge pin (see Illustration 4-2).

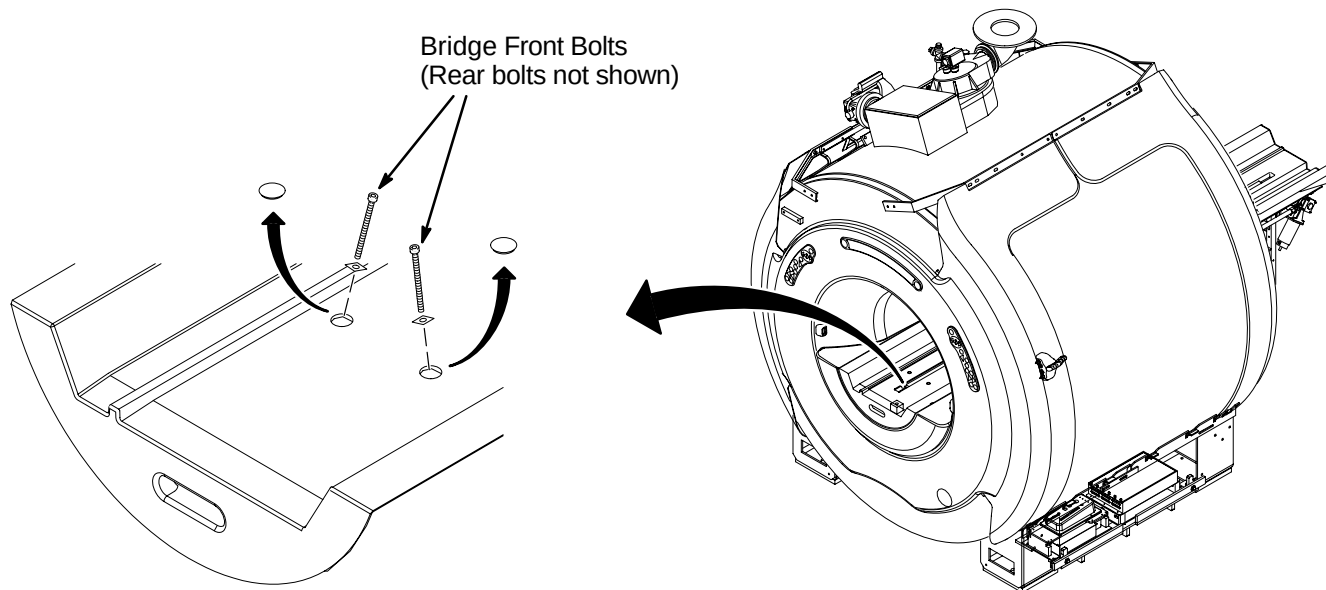
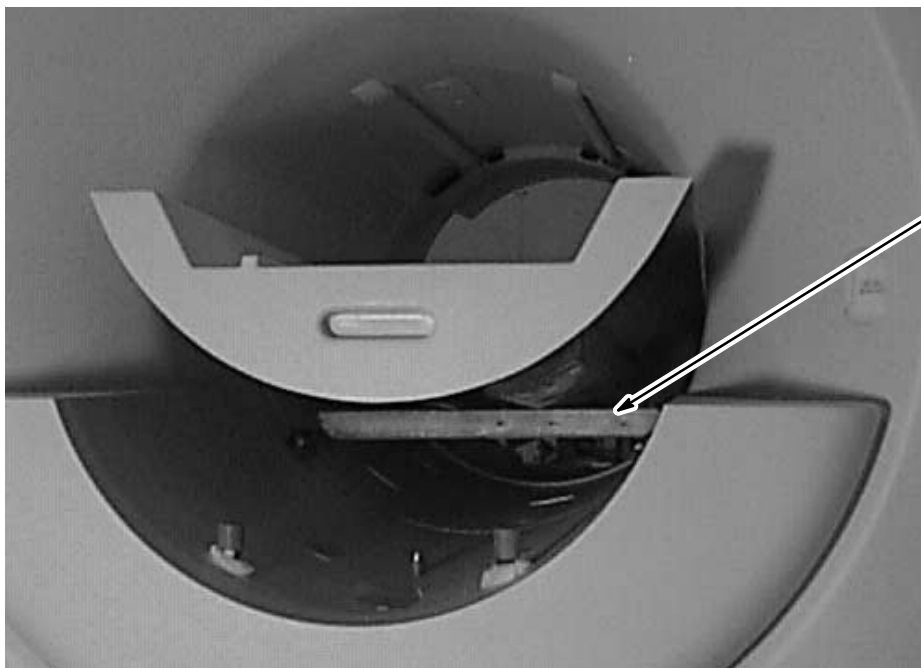
**BRIDGE MOUNTING BOLTS**

ILLUSTRATION 4-2

5. Carefully lift the front end of the bridge and slide wood under bridge as shown in Illustration 4-3. Gently lower the bridge onto the piece of wood.



Place wood to support the bridge.
The wood must be on the RF Coil.

PLACEMENT OF WOOD TO SUPPORT BRIDGE

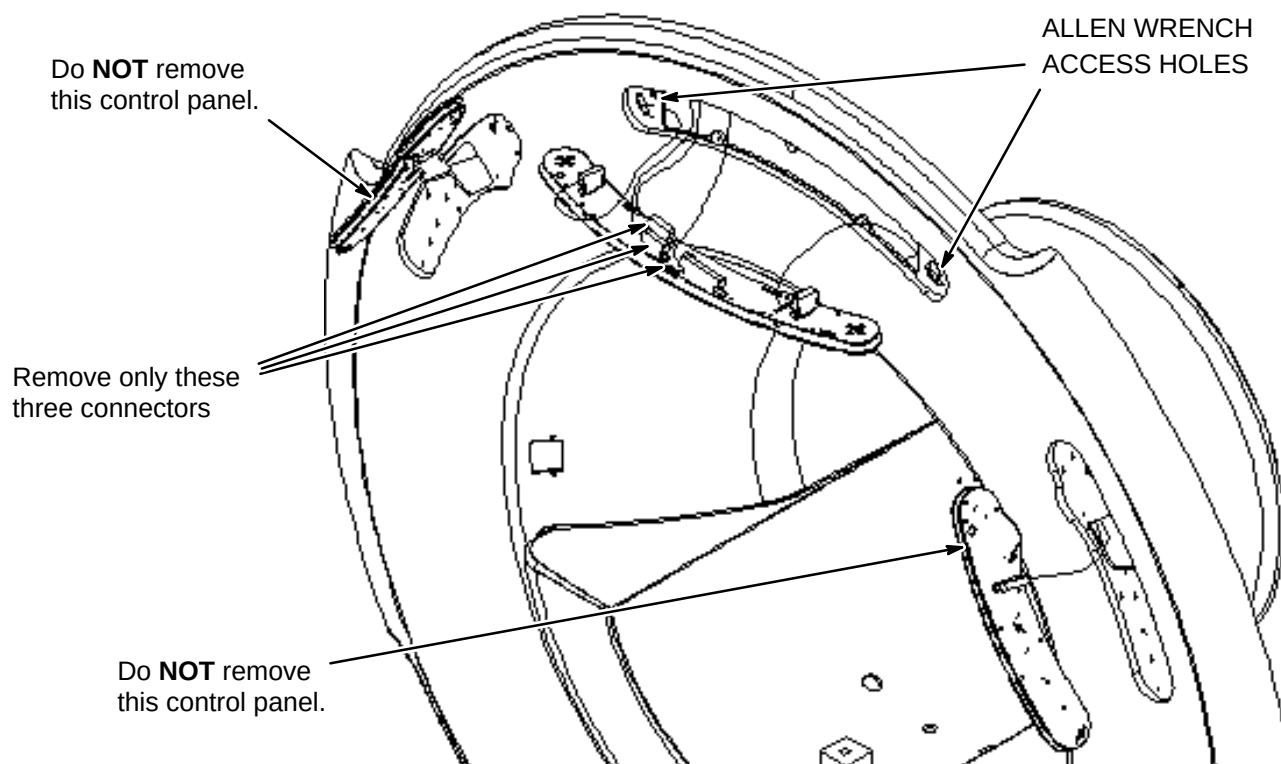
ILLUSTRATION 4-3

4-5 END BELL CONTROL PANEL REMOVAL

Note

There are three control panels on the front end bell. Only the top center control panel is removed for this procedure.

1. The top center control panel is secured to the end bell by one of two methods, which are described in the following paragraphs. Visually inspect the end bell you are removing to determine which method is used.
 - If the central control panel is secured with push-on fasteners, use your fingers to gently pry the panel off the end bell.
 - If the central control panel is secured by two half-turn latches that are accessed with a 3mm allen wrench, insert the allen wrench into each access hole and turn counterclockwise $\frac{1}{2}$ turn to release the control panel as shown in Illustration 4-4.



CONTROL PANEL LOCATIONS (*Earlier revision end bells*)
ILLUSTRATION 4-4

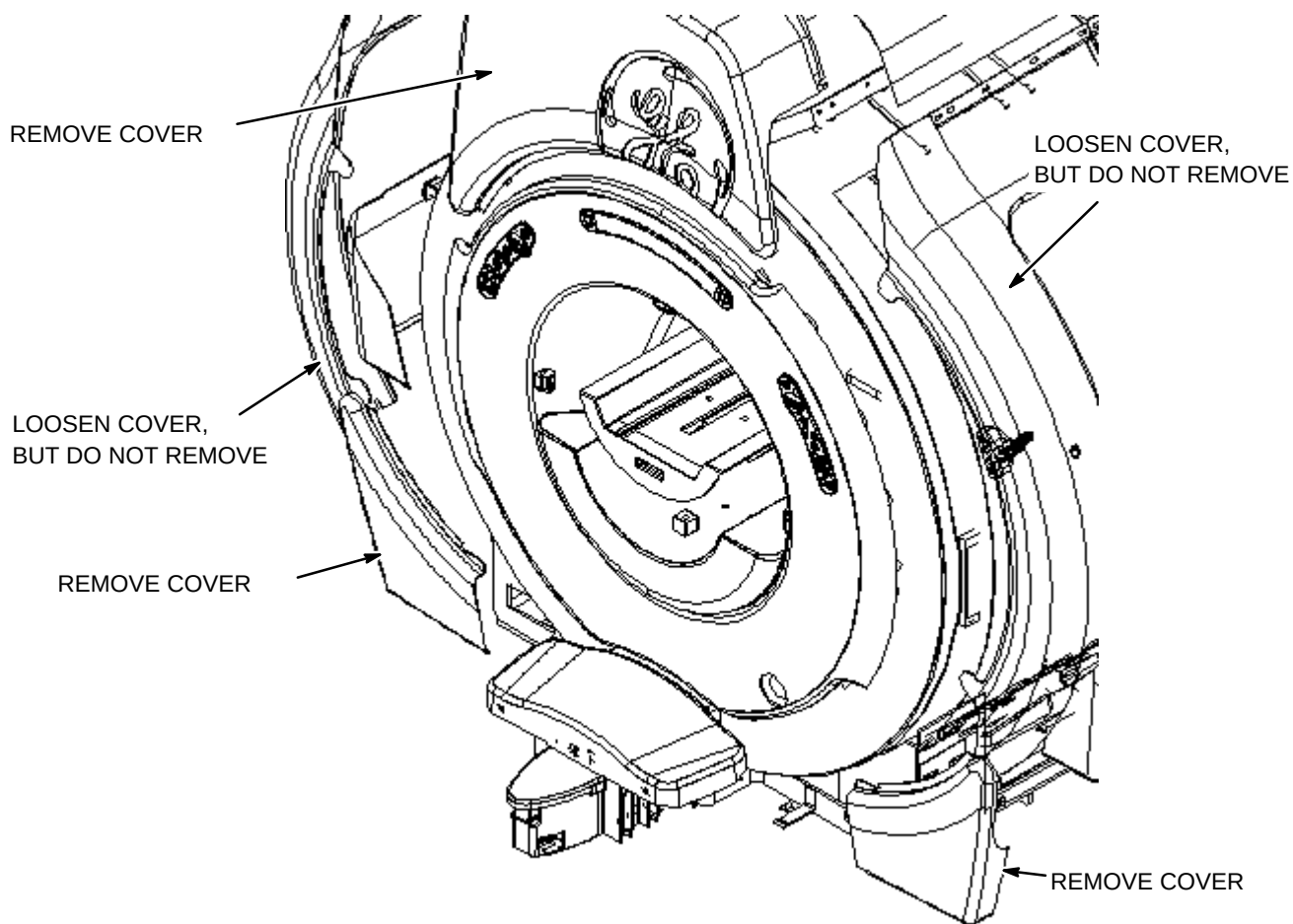
2. Lift the center control panel out and disconnect the three SCSI type II connectors shown in Illustration 4-4. It is not necessary to disconnect the other two connectors.

4-6 END BELL REMOVAL

1. Disconnect the following cables:
 - J10 at the SRI module
 - J20 at the penetration panel (magnet room side)
 - J9 at the patient communication box
2. Remove the front enclosure covers. See Illustration 4-5.

Note

For the two large side enclosure covers, it is not necessary to remove all five securing bolts for each cover. For each of these covers, remove the three bolts closest to the front, and loosen the two rear bolts several turns. This allows enough play to gently move the covers to allow removing and installing the end bell.



WideOpen MAGNET ENCLOSURE, FRONT VIEW
ILLUSTRATION 4-5

4-6 END BELL REMOVAL (Cont'd)

3. Disconnect two latches connecting front end bell and RF tube. See Illustration 4-6.
4. Remove M5 allen cap screw and phenolic spacer. See Illustration 4-6.

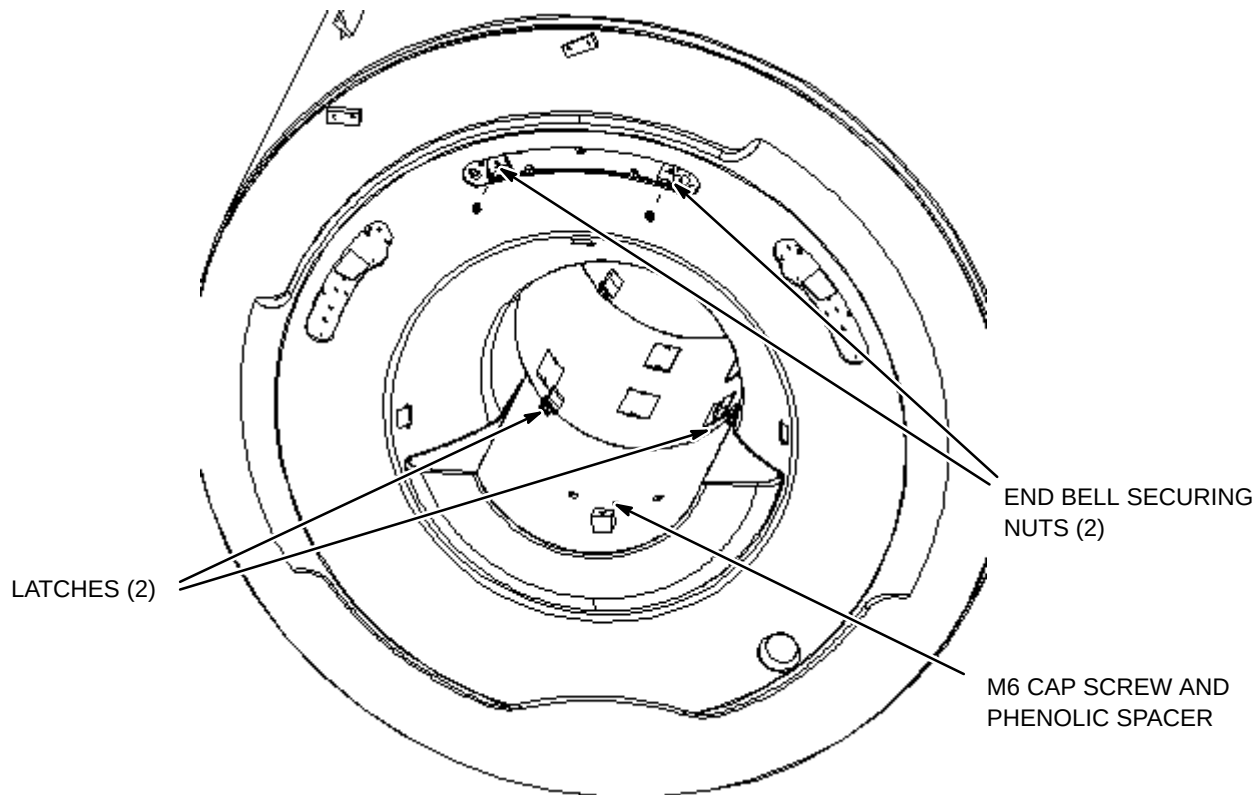
**END BELL, FRONT VIEW**

ILLUSTRATION 4-6

5. Remove two 17mm nuts securing front end bell. See illustration 4-6.
6. Remove front end bell by pulling top out past mounting studs, then lifting vertically to clear side panels and end bell inserts, then pulling out the rest of the way.

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

4-7 SENSOR MOUNTING

1. Remove the end bell from the magnet room.
2. Cut the cable ties securing the existing thermal sensor cable to the end bell.
3. Cut the thermal sensor cable about two feet away from where it exits the thermal blanket.
4. Pull back the thermal blanket to expose the thermal sensor.
5. Remove the bracket directly under the thermal sensor by removing the M3 allen cap screw securing the bracket to the end bell.



Possible equipment damage! Be careful not to damage the end bell when using the hammer and chisel.

6. Use the hammer and chisel to carefully remove the adhesive that secures the sensor to the back side (magnet side) of the end bell. Use a 6 mm diameter socket-head capscrew or bolt to punch out the existing thermal sensor. See Illustration 4-7. (This hole will be re-used for one of the new thermal sensors.)



PUNCHING OUT THE OLD SENSOR
ILLUSTRATION 4-7

7. Refer to Illustration 4-8 for the location of the second hole. Use the permanent marker to mark the location for the second sensor hole. Drill the new hole using a 7 mm or 5/16 inch drill bit. Drill out the old hole using the same bit.

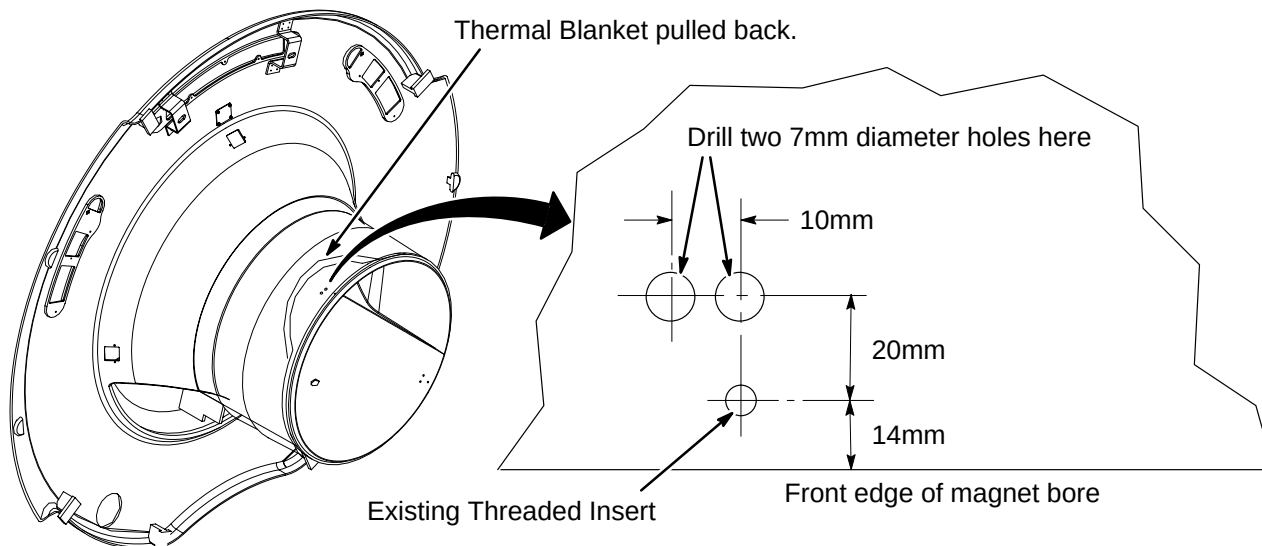
4-7 SENSOR MOUNTING (Cont'd)**SENSOR LOCATION, WideOpen MAGNET**

ILLUSTRATION 4-8

8. Use electrical tape or duct tape to fasten the Dual-Head sensor cable to the old (cut) thermal sensor cable in preparation for pulling the new cable under the thermal blanket. See Illustration 4-9.

NOTE

Be aware that there is a flap of thermal blanket that the sensor can get caught on. Place your hand under the blanket and guide the sensor past this obstacle.

**ROUTING THE REPLACEMENT CABLE**

ILLUSTRATION 4-9

9. Carefully pull the new cable through the route used by the old thermal sensor cable.

4-7 SENSOR MOUNTING (Cont'd)

10. Place a piece of electrical tape or duct tape on the patient side of the end bell to cover both holes. Masking tape may be used if necessary. See Illustration 4-10.

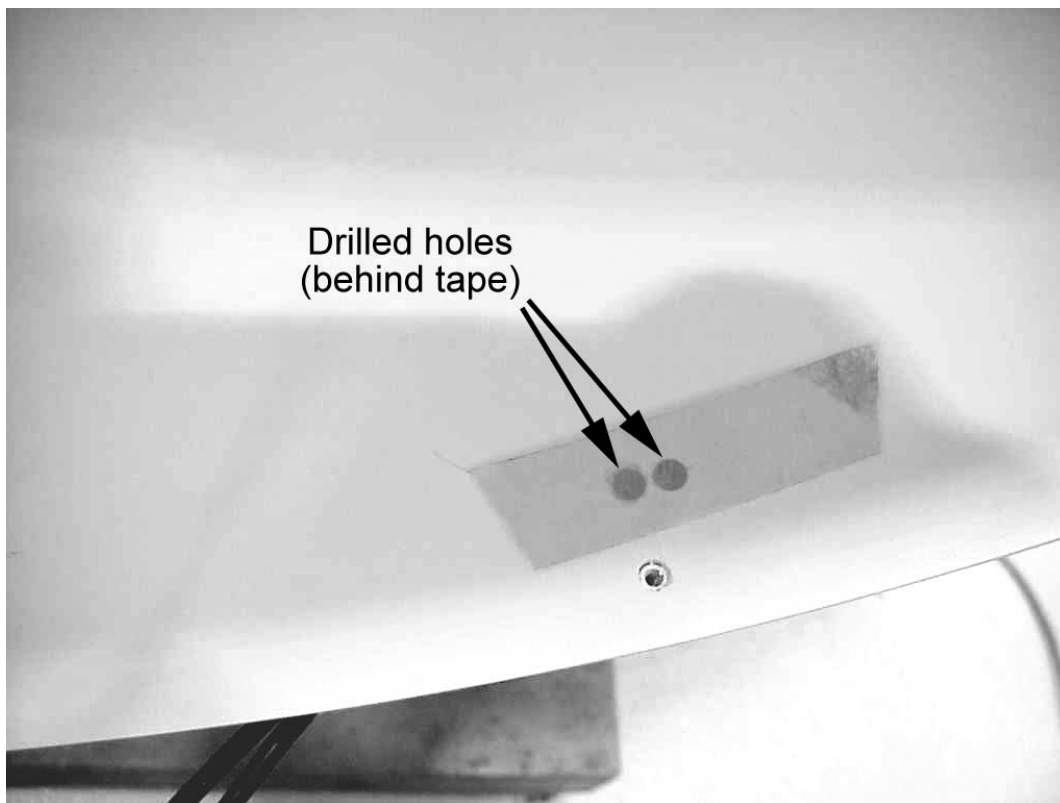
**TAPING OVER THE HOLES**

ILLUSTRATION 4-10

11. Place a moderate amount of epoxy adhesive in both holes.
12. Install the two new sensors into the holes so that the sensors are flush with the tape. Do not allow the sensors to protrude past the finished surface (patient side) of the end bell.
13. Add extra epoxy adhesive at the back of each sensor to ensure that it is securely mounted.
14. Allow time for the epoxy adhesive to set, then gently pull the cables to remove the slack.

Note

The cables must lay flat when pulled taut. This is critical to allow the end bell to fit back on.

15. Use cable ties to secure the thermal sensor cable to the end bell.

4-8 REASSEMBLY OF MAGNET ENCLOSURE

4-8-1 Replacing the Magnet End Bell

Reverse the Magnet End Bell Removal procedure (Section 4-5) to reinstall the magnet end bell. Be sure to follow the following gap specifications:

- All enclosure component seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35 mm). Exceptions may be made in areas where esthetic visual enhancement of the component gap is required by industrial design.
- The gap at the seam between the enclosure and RF coil shall not exceed 0.125 inches (3.175 mm).
- Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

4-8-2 Replacing the Control Panel

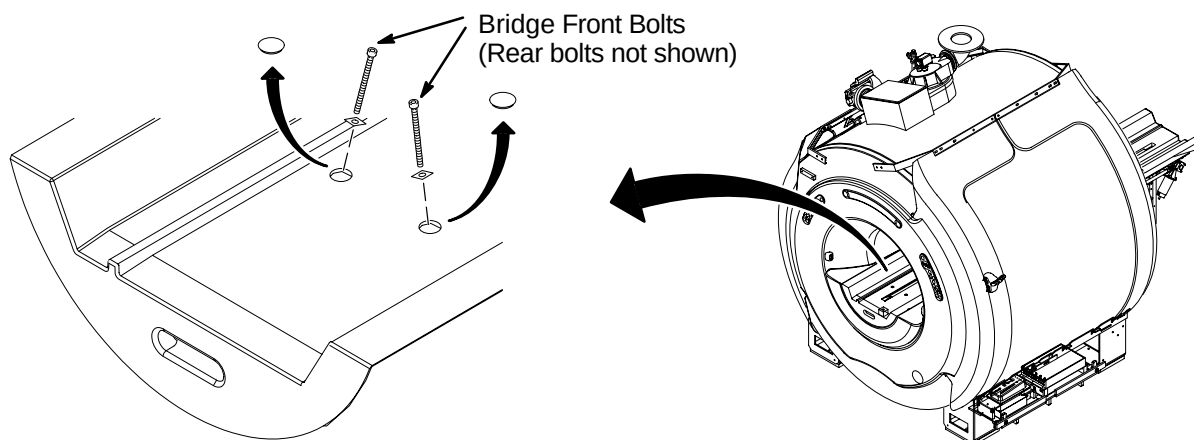


Possible equipment damage! Be sure to connect the Control Panel cables to the correct connectors!

1. Connect the SCSI connectors to the control panel circuit board. .
2. Secure the control panel to the end bell.

4-8-3 Bridge Replacement

1. Remove the wood that was supporting the bridge. Position the bridge for mounting, but do not install the mounting bolts yet.
2. Install the covers removed or loosened in Section 4-6. Where applicable, check for proper fit at edges of end bell.
3. Install the four bolts securing the bridge. See Illustration 4-11.



FRONT BRIDGE BOLTS

ILLUSTRATION 4-11

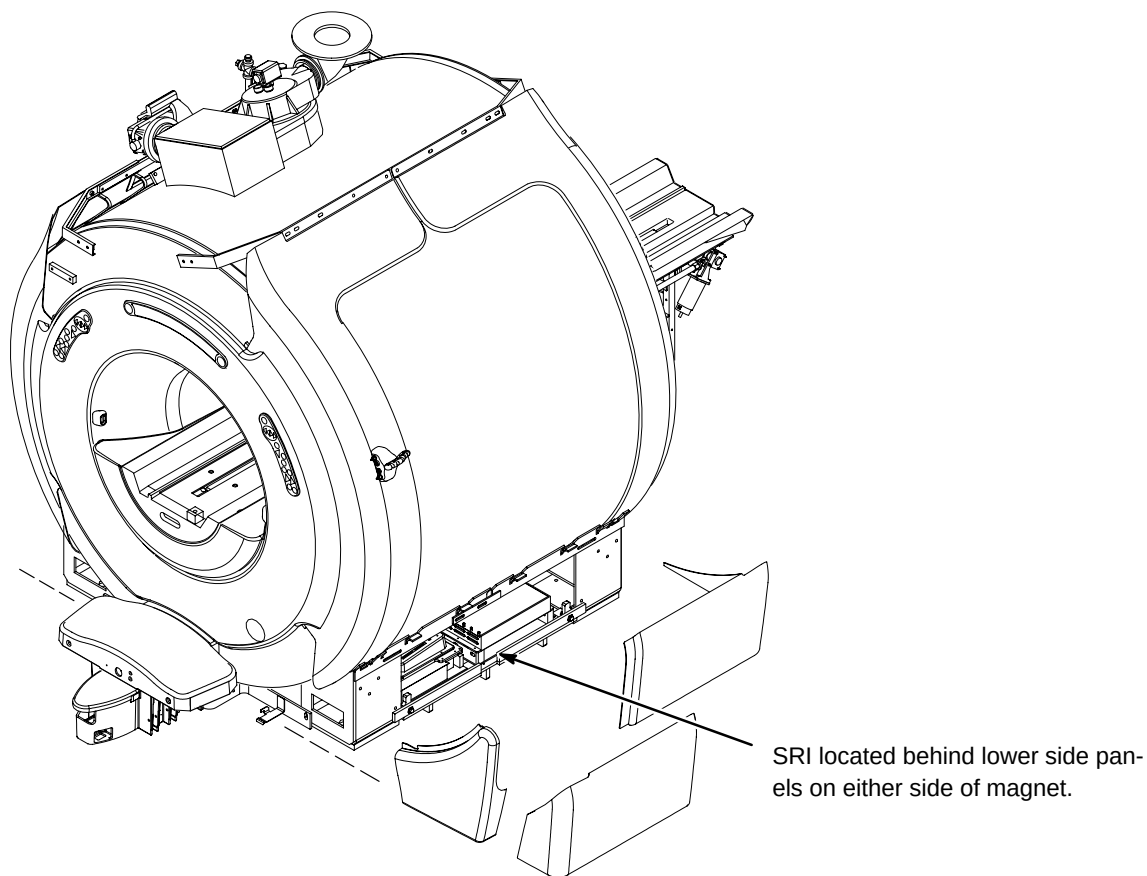
4. Push down the tension arm to apply tension to carriage drive belt (see Illustration 4-1).
5. Install covers on rear pedestal.

4-9 MODIFYING THE SRI MODULE

This section contains complete instructions for replacing the circuit board in the SRI module. The SRI module must be removed from the magnet room before replacing the circuit board. Separate instructions are provided for the WideOpen enclosure magnet and the standard (Horizon) magnet.

4-9-1 Removing the SRI Module

1. The SRI can be located under either the right or left side of the magnet enclosure. Identify which side the SRI is on by locating the patient leads; the SRI will be on that side. See Illustration 4-12.
2. Remove the lower enclosure trim covers to expose the tray that holds the PAC and SRI.



SRI MODULE LOCATION

ILLUSTRATION 4-12

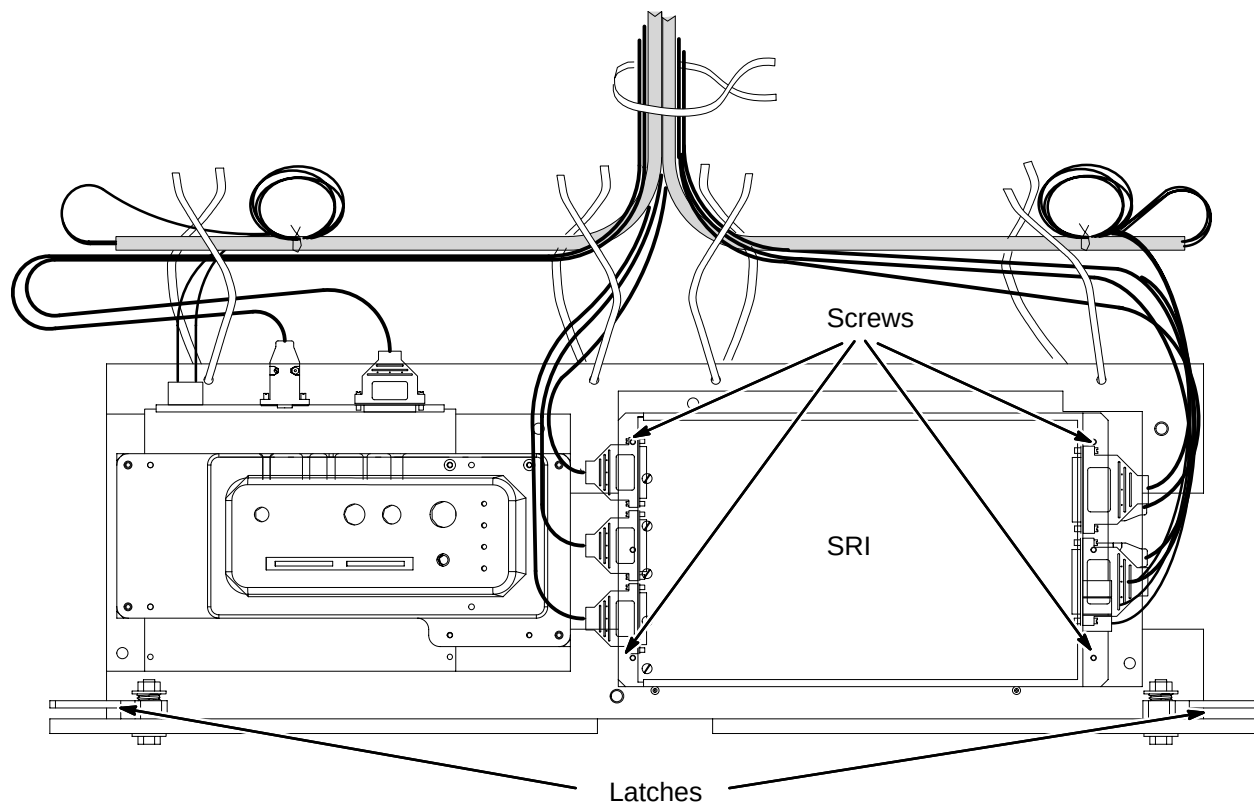
3. Turn latches in holding tray in place. Slide tray out far enough to gain access to the SRI cables. See Illustration 4-13.

4-9-1 Removing the SRI Module (Cont'd)

4. If copper tape is present on the SRI module cover, note all the locations where the tape was applied, then remove the tape.
5. Loosen the six captive screws securing the top cover of the SRI module enclosure, then slide the cover off.
6. Carefully release the three fiber optic cables from their connectors on the SRI PCB and pull the cables free from the SRI module.
7. Disconnect the remaining cables attached to the SRI.
8. Remove four screws holding SRI to tray. See Illustration 4-13.
9. Lift SRI off tray and remove.



Ferrous material present! Remove the SRI module from the magnet room before replacing the SRI module circuit board.



PAC/SRI TRAY
ILLUSTRATION 4-13

10. Replace the existing SRI module circuit board with the replacement circuit board. Refer to "Replacing the SRI Module Circuit Board" in this procedure.

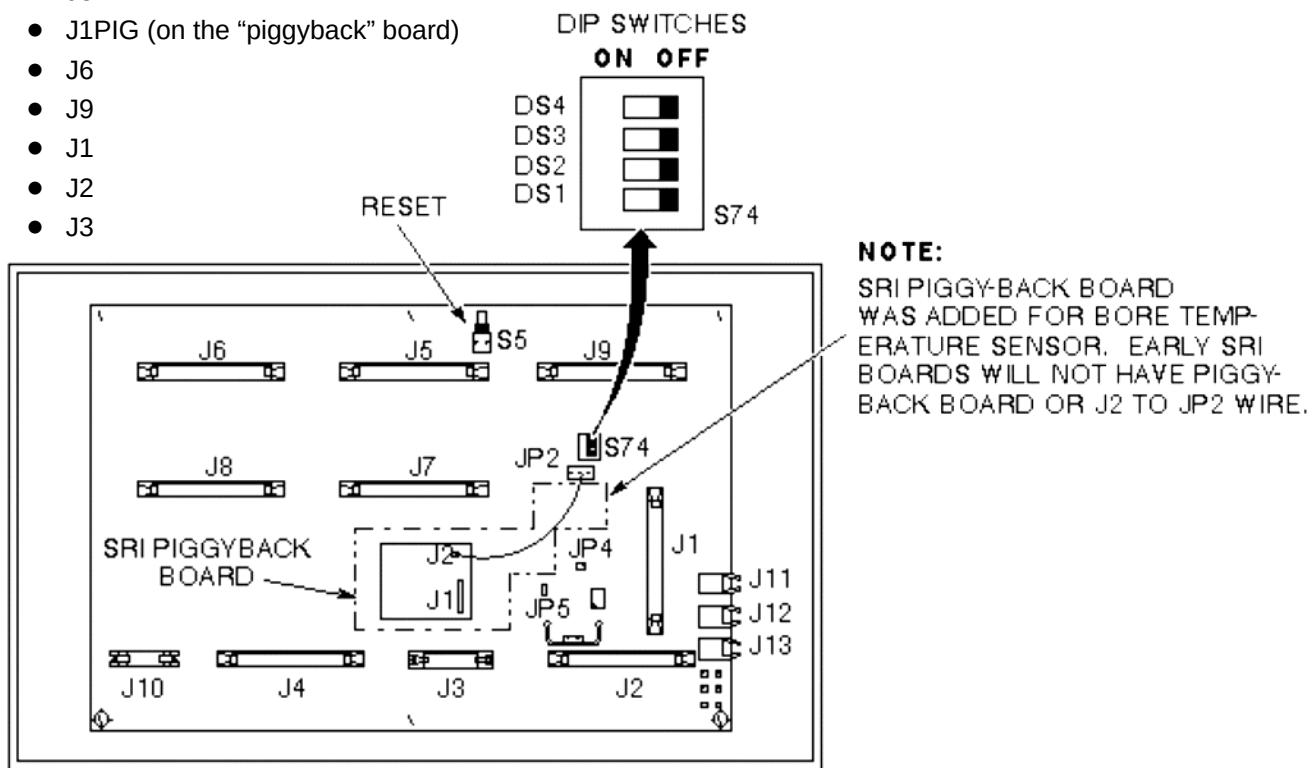
4-9-2 Replacing the SRI Module Circuit Board



Potential equipment damage! Wear a static-grounding wrist strap to prevent ESD (electrostatic discharge) damage to the circuit board.

1. Disconnect all ribbon cables from the SRI circuit board in the following sequence:

- J5
- J7
- J4
- J8
- J1PIG (on the “piggyback” board)
- J6
- J9
- J1
- J2
- J3



SRI BOARD CONNECTION LOCATIONS

ILLUSTRATION 4-14

2. Make sure the free ends of the ribbon cables will not interfere with the removal of the circuit board. If necessary, use masking tape to temporarily secure the free ends of the cables to the outside of the SRI module enclosure.
3. Remove six hex nuts and six washers securing the SRI circuit board to the SRI module enclosure mounting studs.
4. Carefully lift the board off the studs (w/ standoffs) and remove it from the SRI module enclosure. Place the board on an anti-static mat. Do not remove the fish paper underneath the board.
5. Position the replacement board over the six mounting studs.
6. Secure the board using the six hex nuts and six washers removed in step 5
7. Connect the ribbon cables to the board in the reverse sequence listed in step 3.

4-9-2 Replacing the SRI Module Circuit Board (Cont'd)

8. Refer to Tables 4-1 and 4-2 to verify proper jumper and switch settings on replacement SRI board.

TABLE 4-1
SRI BOARD SWITCH INFORMATION

Switch	Schematic Location	Position	Description	Notes
S5	2-B4	Open	BOARD RESET	
S74-SW1	2-H9	Open	(TEST USE)	
S74-SW2	2-H9	Open	(TEST USE)	
S74-SW3	2-H9	Open	(TEST USE)	
S74-SW4	2-H9	Open	(TEST USE)	

TABLE 4-2
SRI BOARD JUMPER INFORMATION

Jumper	Schem. Location	Position	Description	Notes
JP2	2-F6	B	POLARITY SOURCE (SRI without Piggy-back board)	Selects source for LCDPOLAR signal. Position A selects the 8OC196 as source; position B selects off-board source.
JP2	2-C5	Default-Installed	XTAL TEST	Provided for testability
JP5	2-E9	Default-Installed	MEMORY SELECT	Configures memory accessibility. Jumper not installed allows stand-alone diagnostics shell to be accessed. Jumper installed allows normal operation.

4-9-3 Installing The Modified SRI Module

1. Place new SRI module on tray.
2. Secure SRI module to tray using four mounting screws.
3. Connect all previously removed cables to SRI module.

Note

The replacement SRI circuit board may have small protective plugs in the fiber optic connectors. If so, remove the plugs before connecting the fiber optic cables.

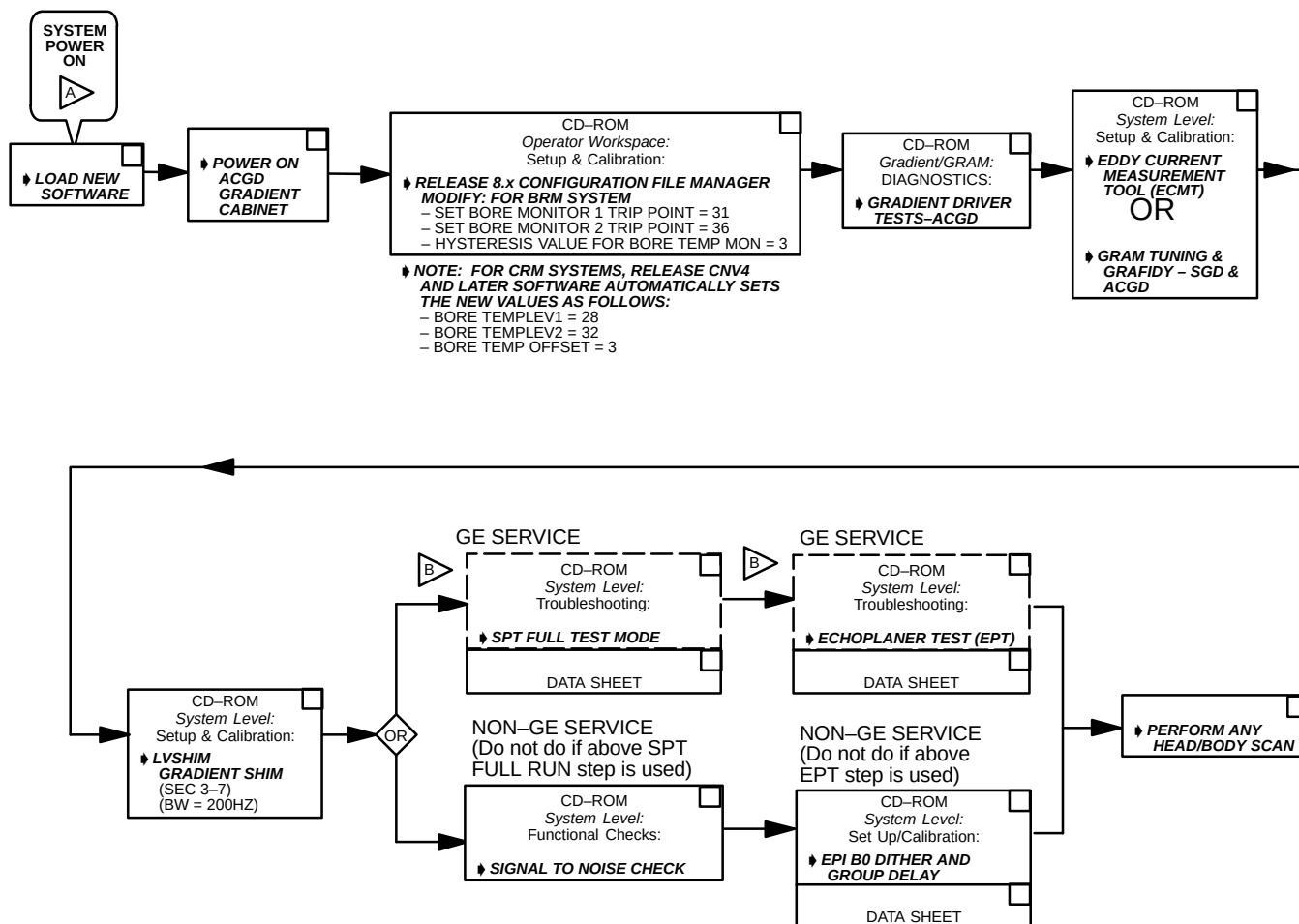
4. Replace the SRI module cover and secure it with the captive screws.
5. If copper tape was removed from the SRI module for this procedure, replace it with new copper tape in the same locations.
6. Slide tray under magnet and secure latches.
7. Replace enclosure trim panels.

4-10 RESTORING POWER

1. Refer to Overview, Section 1-7, POWER ON, for procedures on restoring power.

SECTION 5 – CALIBRATION PROCESS

5-1 CALIBRATION FLOWCHART



A DO NOT TURN ON THE **ACGD** CABINET CIRCUIT BREAKERS (BREAKERS ON FRONT BOTTOM OF CABINET) UNTIL ROOM HUMIDITY IS AT 75% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE IN HIGH HUMIDITY.

B INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.